PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

Smart Classroom and Timetable Schedular

A PROJECT REPORT

Submitted by

SPANDANA KG-20221CSE0359

MISBA AIN- 20221CSE0256

Under the guidance of,

Mr. Syed Mohsin Abbasi

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report " Smart Classroom and Timetable Schedular " is a bonafide work of " Spandana K G(20221CSE0359) and Misba Ain(20221CSE256)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.

Mr. Syed Mohsin Abbasi
Project Guide
PSCS
Presidency University

Dr. Jayavadivel Ravi
Program Project Coordinator
PSCS
Presidency University

Dr. Sampath A K
School Project Coordinators
PSCS
Presidency University

Dr. Asif Mohamed H B
Head of the Department PSCS
Presidency University

Dr. N. Duraipandian
Dean
PSCS
Presidency University

Name and Signature of the Examiners:

Sl. no.	Name	Signature	Date
1			

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We the students of final year B. Tech in COMPUTER SCIENCE AND ENGINEERING, at Presidency University, Bengaluru, named, hereby declare that the project work titled **"**Smart Classroom and Timetable Schedular**"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B. Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

SPANDANA KG	USN: 20221CSE0359	----------
MISBA AIN	USN: 20221CSE0256	----------

PLACE: BENGALURU

DATE: APRIL 2026

ACKNOWLEDGEMENT

For completing this project work, We/I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. We extend our gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the project.

I would like to sincerely thank my internal guide **Mr. Syed Mohsin Abbasi, Assistant Professor,** Presidency School of Computer Science and Engineering, Presidency University, for his moral support, motivation, timely guidance and encouragement provided to us during the period of our project work.

I am also thankful to **Dr. Asif Mohamed H B, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

We express our cordial thanks to **Dr. Duraipandian N**, Dean PSCS, Presidency School of computer Science and Engineering and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my project work.

We are grateful to **Mr. Syed Mohsin Abbasi, and Ms. Shweta Singh, PSCS** Project Coordinators**, Dr. Jayavadivel Ravi, Program Project Coordinator**, Presidency School of Computer Science and Engineering, or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

We are also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

SPANDANA KG

MISBAAIN

Abstract

Smart Classroom and Timetable Schedular is a clever and almost self, operating timetable scheduling program aimed to tackle the complexity and slow work of traditional academic timetable generation. Timetable scheduling represents a key administration workload at educational institutions like schools, colleges, and universities, as it involves coordinating the availability of faculty, the requirements of different subjects, the allocation of classrooms, the working hours, and the constraints imposed by the institution. If done manually, this whole operation takes a lot of time, is prone to errors, and becomes almost uncontrollable at the occurrence of last, minute changes.

The main goal of Smart Classroom and Timetable Schedular is to smartly automate timetable generation; in fact, the solution of automation provided by this platform is capable enough to assure not only accuracy and flexibility but also a very user, friendly interface. In the past, manual labour, spreadsheet work, and trial, and, error strategies have been the basis of conventional scheduling which usually results in various conflicts such as faculty clashes or a double, booked classroom, inefficiently used resources, and the unfair distribution of faculty workload. Moreover, changes in the schedules brought about by events that cannot be planned beforehand, such as faculty leave or substitution of subjects, would entail a great deal of reworking.

Smart Classroom and Timetable Schedular effectively tackles the above challenges by auto, producing well, optimized and conflict, free academic schedules while at the same time staying strictly within the limits set by ready, made academic constraints. The system takes inputs that are well, structured and consequently, there is data regarding faculty, subject allocation, and time session availability, classroom resources, and institutional rules. Apart from ensuring that there are no faculty members who have been given overlapping classes thus facilitating proper utilization of classrooms and subjects being efficiently allocated with the required number of sessions, with the help of an optimized scheduling algorithm, the system accomplishes all these with an extra layer of automation.

Table of Content

Sl. No.	Title	Page No.
	Declaration	iii
	Acknowledgement	iv
	Abstract	v
	List of Figures	viii
	List of Tables	ix
	Abbreviations	x
1	Introduction	
	1.1 Background	1
	1.2 Statistics of project	1
	1.3 Prior existing technologies	2
	1.4 Proposed approach	3
	1.5 Objectives	4
	1.6 SDGs	5
	1.7 Overview of project report	6
2	Literature review	8
3	Methodology	12
4	Project management	
	4.1 Project timeline	17
	4.2 Risk analysis	18
	4.3 Project budget	19
5	Analysis and Design	
	5.1 Requirements	20
	5.2 Block Diagram	20
	5.3 System Flow Chart	21
	5.4 Choosing devices	23
	5.5 Designing units	23
	5.6 Standards	25
	5.7 Mapping with IoTWF reference model layers	25

	5.8 Domain model specification	26
	5.9 Communication model	28
	5.10 IoT deployment level	30
	5.11 Functional view	32
	5.12 Mapping IoT deployment level with functional view	35
	5.13 Operational view	39
	5.14 Other Design	42
6	Hardware, Software and Simulation	
	6.1 Hardware	45
	6.2 Software development tools	50
	6.3 Software code	53
	6.4 Simulation	56
7	Evaluation and Results	
	7.1 Test points	58
	7.2 Test plan	62
	7.3 Test result	66
	7.4 Insights	67
8	Social, Legal, Ethical, Sustainability and Safety Aspects	
	8.1 Social aspects	69
	8.2 Legal aspects	71
	8.3 Ethical aspects	74
	8.4 Sustainability aspects	76
	8.5 Safety aspects	77
9	Conclusion	80
	References	81
	Base Paper	82
	Appendix	83

List of Figures

Figure ID	Figure Caption	Page No.
3.1	Smart Classroom and Timetable Schedular Development Process	14
3.2	Agile Scrum Journey	16
4.1	Project TimeLine	17
5.1	Block Diagram	21
5.2	System Flow Chart	22
5.3	Functional View of Smart Classroom and Timetable Schedular	35
5.4	Mapping of IoT Deployment Level with Functional View	38
5.5	Operational View of Smart Classroom and Timetable Schedular	41
5.6	Other Design Overview of Smart Classroom and Timetable Schedular	44
6.1	Server-Side Hardware Requirements	49
6.2	Client-Side Hardware Requirements	52
7.1	Testing Process Flow	62
7.2	Timetable Evaluation Flow	66
8.1	Ethical and Legal Compliance Flow	71
8.2	Safety and Sustainability Flow	74

List of Tables

Table ID	Table Caption	Page No.
4.1	Risk Analysis	18
5.1	Choosing Device	23
5.2	Functional View of Smart Classroom and Timetable Schedular	34
5.3	Mapping IoT Deployment Level with Functional View	38
5.4	Operational View of Smart Classroom and Timetable Schedular	41
5.5	Other Design Considerations in Smart Classroom and Timetable Schedular	44
6.1	Simulation Test Scenarios	57
7.1	Test Points and Results	61
7.2	Test Plan Overview	65
8.1	Social, Legal and Ethical Aspects	70
8.2	Sustainability and Safety Considerations	73

Abbreviations

Abbreviation	Full Form
API	Application Programming Interface
CDN	Content Delivery Network
CSP	Constraint Satisfaction Problems
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DB	Database
HDD	Hard Disk Drive
HTML	HyperText Markup Language
HTTP/HTTPS	Hypertext Transfer Protocol / Secure
IDE	Integrated Development Environment
IoT	Internet of Things
IoTWF	Internet of Things World Forum
JSP	JavaServer Pages
LAN	Local Area Network
NP	Nondeterministic Polynomial time (Complexity)
QR	Quick Response
RAM	Random Access Memory
RFID	Radio Frequency Identification
SDG	Sustainable Development Goal
SSD	Solid State Drive
UI	User Interface
UPS	Uninterruptible Power Supply
URL	Uniform Resource Locator
USN	University Seat Number

Chapter 1

Introduction

1.1 Background

In schools, a timetable is the process of deciding which courses, teachers, and time slots will be combined so that there are no conflicting activities and stakeholders' needs will be met. Manual work, the mainstay of the traditional methods, is usually the first source of conflicts such as double booking, uneven load distribution, and under-use of the resources.

As scheduling problems are of combinatorial nature, this task is considered to be NP, complete in terms of algorithmic complexity. Hence it needs to be systematically solved with computational methods when dealing with large datasets or dynamically changing environments. Several research works have investigated automation of this process through optimization techniques such as evolutionary algorithms, constraint satisfaction, and decision, support systems.

Our project Smart Classroom and Timetable Schedular is an illustrative example of the application of a constraint, based heuristic algorithm that iteratively generates timetable options while at the same time checking for hard constraints such as no faculty or room overlap and at the same time optimizing soft objectives like balancing workload and classroom utilization. The design of the system allows for multi, level security and easy management of academic entities.

1.2 Statistics of Project

Studies and surveys in academic administration show that: More than 60, 70% of educational institutions still rely on manual or semi, automated tools such as spreadsheets for creating timetables. Depending on the size of the institution, manual timetable preparation may take several days or even weeks. About 30, 40% of scheduling errors are due to human factors such as the oversight of clashes of faculty or the double booking of rooms. After the implementation of automated scheduling systems, institutions have experienced a 50, 70% decrease in the amount of time spent in making timetables.

Smart Classroom and Timetable Schedular focuses on resolving these problems by dramatically lowering scheduling time, totally removing conflicts, and increasing efficiency. The initiative shows how automation can simplify academic planning and ease the burden of administration.

1.3 Prior Existing Technologies

There are many timetable scheduling solutions and tools today, including manual scheduling systems, spreadsheet tools, rule, based systems, and optimization, based approaches. Manual scheduling refers to the use of paper, based planning or very basic spreadsheet tools only, thereby it being very time, consuming, error, prone and hard to update. Spreadsheet, based tools such as Microsoft Excel can provide some help but are not equipped with intelligent constraint handling and thus, require a lot of manual intervention.

Rule, based scheduling systems ensure better precision by basing the decisions on the rules defined by the users; nonetheless, if the requirements increase, these systems become less flexible and scalable.

Some more sophisticated optimization, based systems turn to methods such as Genetic Algorithms, Constraint Satisfaction Problems (CSP), or heuristic, based procedures for timetable generation. These methods, being highly efficient, also come with the drawbacks of being complicated, quite hard to set up, and not very user, friendly for administrators in the real, world situations. Consequently, many existing systems are not able to achieve the right mix of usability, flexibility, and efficiency. This deficiency in the market served as a stimulus for the Smart Classroom and Timetable Schedular development.

1.4 Proposed Approach

The proposed solution of Smart Classroom and Timetable Scheduler system is largely based on the disuse of manpower and improvising the educational timetable generating techniques by stepwise data input, scheduling logic with the awareness, and timetable handling suitably. The same is capable of solving the issues of the classical manual ways of scheduling by rendering more efficiency accuracy scalability, and the ability to change with the ever-changing academic environment. Instead of doing the planning manually or with the spreadsheet, Smart Classroom and Timetable Scheduler bring in an automated system that features consistency, reliability, and easy administration as the leading factors. Smart Classroom and Timetable Scheduler adopts a step by step approach for timetable making.

In the old days, the teachers used to do the planning with a pen and paper or using some static tools but now the system makes different timetables based on the user's constraints and the institution's rules. The main point of this scheme is to make timetable creation a data, oriented activity where scheduling decisions are made logically through automated rules. By doing this, it is ensured that scheduling is conflict free, resource is used to the maximum, human intervention is at a minimum, and the issue of maintenance is dealt with. On the other hand, the modular setup of the system's parts allow each element to work independently and can be replaced or upgraded independently without affecting the overall system.

With the help of a well, designed interface, the system makes it possible for the administrators and academic coordinators to have a glance, handle, and make changes to the generated timetables without any difficulty. Within this interface, the users can make amendments to input data, keep an eye on the distribution of faculty workload, and print or save timetables for the use of the institution. The user interface is so intuitive and straightforward that even nontechnical users will be able to access without the need of any support, thus boosting the acceptance and ease of use of the system.

1.5 Objectives

The main goal of Smart Classroom and Timetable Schedular project is to come up with a smart and automated timetable scheduling system capable of generating academic timetables easily. The system is expected to be a totally new and hugely efficient method of operation by providing a stable digital solution that is superior in every aspect to traditional manual and semi, automated methods.

Smart Classroom and Timetable Schedular is set up to be one of the best timetable scheduling systems by automating the creation of timetables almost without the need for any human input. Human scheduling is not only slow, but it usually contains errors such as faculty clashes, overlapping classes, and room misallocation that can be avoided. The system acting as the agent of human scheduling will be able to prepare timetables in a very short time and at the same time results of scheduling decisions will be always correct and reliable.

Further, it is a big win to have a system that is able to fix faculty and student scheduling conflicts and at the same time helps the school to use its resources very well. Smart Classroom and Timetable Schedular helps faculty not to have any class scheduling conflicts and it also helps to avoid the double booking of a classroom. Moreover, the system guarantees that a subject is assigned the required number of sessions and the classrooms at the school are made best use of. Achieving this target is sure to result in better academic management and resource utilization of the institution. Furthermore, the project aims at designing a teaching timetable in a way that the faculty workload is distributed evenly and each one is being treated fairly.

Smart Classroom and Timetable Schedular, by taking into account the presence of faculties and scheduling their working hours in a way that they are not overloaded, introduce transparency and fairness in the distribution of teaching duties. This makes the teachers more relaxed and content with the whole work. The key aspect of the system is the flexibility and the adaptability. When the admin is confronted with the actual situation of the teacher's unavailability, change of subjects or reassignment of classrooms, Smart Classroom and Timetable Schedular empower the admin to modify the schedule very easily. The feature of changing the schedules without having to start composing a new timetable from the scratch significantly leads to lessening the disruption to the educational process.

1.6 Alignment with Sustainable Development Goals (SDGs)

The Smart Classroom and Timetable Schedular project is connected to several of the United Nations Sustainable Development Goals (SDGs) as it promotes better use of educational resources, new technology, and green ways of doing things. Introducing Smart Classroom and Timetable Schedular as a tool to help teachers and students by making the timetable scheduling process automatic and efficient is a move towards the establishment of smart, green educational institutions. The project through the productive use of technology backs the worldwide idea of inclusive, innovative, and environmentally responsible development.

1.6.1 SDG 4 Quality Education

SDG 4 is focused on providing inclusive and equitable quality education and promoting lifelong learning opportunities for everyone. Smart Classroom and Timetable Schedular is a tool that helps achieve this objective to a great extent.

By making the academic schedule more efficient and reliable, a well, structured and a conflict, free timetable essentially leads to uninterrupted teaching and learning activities. Students are able to attend their classes without confusion or overlap. It also helps the faculty members to deliver their lectures in a more effective way by reducing the headache that comes with scheduling and the imbalance of the workload. Smart Classroom and Timetable Schedular through structured academic planning and better utilization of institutional resources not only supports the proper functioning of the whole educational system but also improves the quality of the education.

1.6.2 SDG 9 Industry, Innovation, and Infrastructure

SDG 9 intends to build resilient infrastructure, promote inclusive and sustainable industrialization, and support innovation. Smart Classroom and Timetable Schedular is directly aligned with this objective as the tool brings automation and intelligent scheduling systems to school administration. The initiative is a great indicator that the use of digital solutions along with next-generation technologies can not only help improve efficiency of operations but can also bring new ideas to the conventional ways and assist in creating a strong institutional framework. Being modular and made up of scalable design elements, Smart Classroom and Timetable Scheduler are leading the digital education transformation and simultaneously, they

recognize the demand for inventive ways for more effective administrative systems.

1.6.3 SDG 12 Responsible Consumption and Production

One of the things SDG 12 aims to is the sustainable management and efficient use of global resources. The Smart Classroom and Timetable Scheduler contributes towards the realization of this goal through the elimination of traditional printing and paper-based methods which were typical of manual timetable planning. The shift towards scheduling digitization significantly lessens the volume of paperwork, curtails the wastage of resources, and even eliminates the need for performing multiple manual corrections. Furthermore, the optimal use of the faculty and classrooms time is another example of how the institution's resources can be used in a responsible way. Powered by automation and digital management, Smart Classroom and Timetable Schedular is a pioneer of educational establishments' green and sustainable operations.

1.7 Overview of the Project

This report focuses on the development of Smart Classroom and Timetable Schedular, a smart, automated timetable scheduler that can greatly aid academic planning and streamline administrative tasks in schools. Besides describing the current scheduling problems, this paper also discusses the suitability and effectiveness of Smart Classroom and Timetable Schedular in delivering an optimized and automated scheduling solution.

Scheduling education activities is first introduced as the main topic of the report followed by the identification of the drawbacks of manual and partially automated scheduling methods. The report notably elucidates the fundamental concepts behind design elements and system specifications of Smart Classroom and Timetable Scheduler.

Besides, it discusses the significance of learning scheduling constraints, school policies, and customer's needs, as ways not only to make the system functional but also to render it effective and user friendly. In fact, the paper proceeds along similar lines by depicting the two phases of the systems - conceptual and design - and how they were turned into a practical timetable by means of extensive data gathering, scheduling constraint logic, and component-based architecture. Apart from that, implementation stages are indicated through flowcharts to

describe how the system carries out data extraction, rule application, and generating conflict-free timetables. Because of features such as the ability to be changed extended easily understood, and the presence of the interface that is not only visually pleasing but also intuitive educational departments are ready to embrace automated systems. Furthermore, testing techniques, system's behavior in different situations, and outcomes are among the ways the paper measures the performance of the Smart Classroom and Timetable Scheduler. The main aim of the study is to demonstrate the potential of automation to assist in drastically reducing errors, saving time, and making efficient use of resources.

Chapter 2

Literature Review

2.1 Introduction

The literature review is a key tool in gaining a clear grasp of the domain and pinpointing the gaps where the proposed idea fits in. It also helps one to find a consensus among authors and to support the views with evidences/examples. This is thoroughly discussed in this paper as well.

Timetable scheduling is an NP, hard problem with multiple constraints and requirements. One of the reasons for attracting researchers to this problem is that it has attained a wide range of real, life applicability and scenarios. These constraints include teacher availability, class, room capacity, subject, weight, and institutional policies.

Some of the most recent works are conducted in this area alongside the main researches that have been carried out in the past. Hence, preliminary manual methods and basic software tools are always reference points for the development of computational methods such as genetic algorithm, neural networks, simulated annealing, and meta, heuristics. While most of the systems reviewed provide substantial performance improvement over manual scheduling, they often show lack of flexibility, scalability or user, friendliness, which has prevented their full utilization in the academic environment.

Furthermore, the literature demonstrates the tendency of the scholars to put emphasis on the importance of the integration of automation and intelligent systems into educational management. The latest trend is to develop intelligent systems that can adapt to different situations within educational management. Systems should be capable of adapting to changed situations, minimizing clashes, and making the best use of resources. This paper surveys the studies concerned with the problem of academic scheduling and compares different approaches to scheduling. Besides that, it provides a basis for Smart Classroom and Timetable Schedular system proposal through the identification of existing gaps.

2.2 Research Papers Reviewed

[1] Burke, E. K., & Petrovic, S. (2002) "Recent Research Directions in Automated Timetabling"

This article provides an in, depth look at various methods for automated timetabling, illustrating the considerable difficulty of scheduling problems, which in essence arise from the constraints that have to be satisfied. To generate feasible timetables, the authors review different kinds of heuristic and optimization, based techniques. The paper underlines constraint managing and adaptability as being two of the key factors, which essentially correspond to the Smart Classroom and Timetable Schedular system guiding principles.

[2] Schaerf, A. (1999) "A Survey of Automated Timetabling"

This work is a review of different automated timetabling methods, which are classified depending on whether they are based on constraint satisfaction or optimization techniques. The major problems that the paper points to are those of faculty clashes and room allocation conflicts in timetable generation. Such results confirm that an automated and constraint, based approach, such as the one implemented in the Smart Classroom and Timetable Schedular project, is necessary.

[3] Burke, E. K., De Causmaecker, P., Berghe, G. V., & Van Landeghem, H. (2004)

"The State of the Art of Nurse Rostering" In an attempt to show scheduling problem similarities between the two domains, by referencing some of the issues addressed in academic timetabling, this study has mainly focuses on nurse rostering and considers those problems such as shift conflicts and workload balance. Besides, the authors emphasize equitable workload distribution and employee flexibility as the solution to such problems. The latter are very relevant for the academic world and Smart Classroom and Timetable Schedular has incorporated them in its faculty workload balancing tool.

[4] Kumar, R., & Singh, M. (2017), "Automated Timetable Generation Using Genetic Algorithm"

The main content of this document is proposing the use of genetic algorithms to create efficient academic timetables. It is shown through the research that an automated system can drastically shorten scheduling time and reduce conflicts versus the manual method. Even though the method is a computationally heavy one, it still, emphasize the fact that automation is necessary for the efficient creation of timetables, thus presents the same rationale behind Smart Classroom and Timetable Schedular.

[5] Shah, P., Patel, D., & Mehta, S. (2019) "Smart Classroom Management System Using Automation"

This publication describes the automation of classroom and timetable management. The authors underline the advantages of using automation such as greater efficiency, less use of paper, and improved inter, departmental coordination. The findings of this work correspond very well with the aim of Smart Classroom and Timetable Schedular, which intends to transform scheduling into a digital and automated process for the purpose of increased institutional productivity.

[6] Ahmed, S., & Rahman, M. (2020) Constraint, Based Scheduling for Academic Timetables

The paper talks about a constraint based scheduling model that is aimed at generating a timetable without any conflict through the use of hard and soft constraints. The study focuses on the emphasis flexibility and adaptability in the real, world academic environments. These ideas have had a direct impact on the constraint, based scheduling logic that has been implemented in the Smart Classroom and Timetable Schedular system.

2.3 Research Gaps Identified

2.3.1 Lack of Flexibility in Existing: Systems Many existing timetable scheduling systems produce static timetables that should be completely regenerated if there are minor changes such as faculty leave or classroom reassignment. Such inflexibility makes the systems unsuitable for real, world academic environments, where a great deal of updating takes place.

2.3.2 High Dependence on Manual Intervention: Many institutions still use manual or semi,

automated tools like spreadsheets, which involve a great deal of human work and result in such approaches being error, prone, inconsistent, and inefficient, especially when the institution grows in size and complexity.

2.3.3 Inadequate Handling of Multiple Constraints: Existing solutions regularly allow only a limited set of academic constraints to be satisfied. Other constraints such as faculty availability, subject requirements, classroom capacity, and institutional policies can be neglected. Therefore, the resulting schedules are conflict, prone and far from optimal.

2.3.4 Poor User Interface and Usability: Several research, based scheduling systems have directed all their efforts to algorithmic complexity while forgetting the user experience. Therefore, administrators and academic coordinators find these systems hard to use, which keeps them out of practical use.

2.3.5 Limited Scalability: Several timetable generation systems produce excellent results on small datasets but transition to failure when an institution with various departments, a large number of faculty members, and a complex scheduling configuration is involved.

Chapter 3
Methodology

3.1 Introduction

This chapter is devoted to the methods used in the design and the development of Smart Classroom and Timetable Schedular, a smart and automated timetable scheduling system. The methodologies describe the broad development strategy, tools, procedures, and workflow that have been used to turn project needs into a working system.

A well, structured development methodology would guide through the phases of planning, control of changes, and improvement of the product while development is still going on. Since software development is an inherently changeable process and there is a need for flexibility, an Agile, based approach was chosen as the best fit for this project.

3.2 Chosen Development Methodology – Agile Scrum

The development team had decided to adopt the Agile Scrum methodology for Smart Classroom and Timetable Schedular as it is iterative, flexible, and is very much human, centered. Agile Scrum is quite different from the conventional linear models and therefore allows for the product to be developed continuously, customers feedback to be obtained regularly and making it possible to respond quickly to changes. Hence, it is a perfect match for the Smart Classroom and Timetable Schedular type of projects in which scheduling logic, constraints, and user requirements could be a matter of change before the product is finally delivered.

With Scrum, the overall project is divided into small, manageable pieces of work, called sprints, where each sprint results in a working feature. Through regular reviews and testing mistakes can be detected early and thus there is a continuous opportunity for incorporating improvements. Besides that, Agile Scrum promotes collaboration among team members, explicitly defines roles, and ensures transparency throughout the entire development process.

3.3 Project Phases: Our Development Journey

The development of Smart Classroom and Timetable Schedular was broken down intentionally into several phases. The first phase was requirement gathering and analysis in which a very thorough investigation of scheduling problems in academics at the different levels as well as the getting limitations of the institution and users' expectations was conducted. The system design stage came next and the project architects outlined the whole plan, divided the paths/data flow, and figured out the scheduling algorithms.

During the coding phase the developer's focus was on creating the basic modules such as data input handling, constraint, based scheduling, timetable generation, and designing and developing the user interface components. Testing was not an independent activity but rather it was done continuously as development progressed to ensure both correctness and stability. Finally, through the deployment and assessment phase, the system was up to date with the use of real situations which confirmed that the solutions offered were in agreement with the project goals.

3.4 Process Flow Diagram

The Smart Classroom and Timetable Schedular process flow starts with an administrator who provides a variety of well, structured input items such as staff details, subjects, classrooms, and time slots. These inputs are checked for correctness and then recorded in the system database. After that, the scheduler engine uses constraint, based logic to produce a conflict, free schedule. The schedule thus created is checked, shown to the user, and saved for future reference. The system only changes those parts of the schedule that have been changed, thereby retaining the flexibility and efficiency if the changes are required.

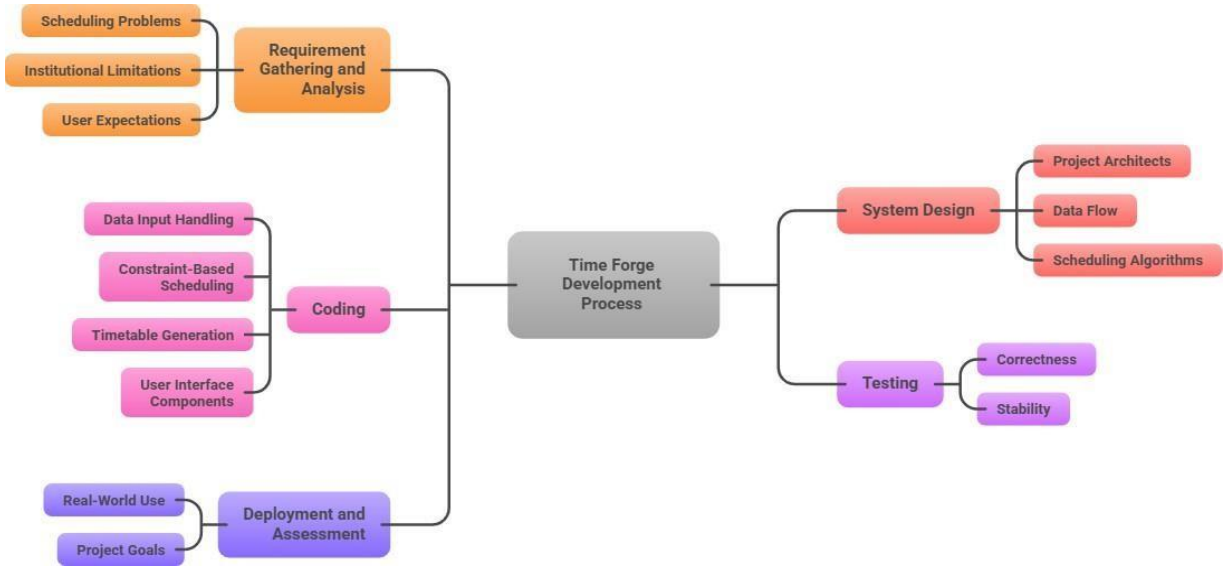

Figure 3.1: Smart Classroom and Timetable Schedular Development Process

3.5 Use of Tools and Technologies

Frontend: Provides an intuitive interface for school admins to effortlessly enter data and visualize the generated timetables.

Styling: Produces neat, responsive, and highly user, friendly design which makes navigation and usage very simple.

Language: Develops backend operations and timetable scheduling algorithms which can take the given data and generate timetables.

Database: Faculty, subject, classroom, time slot, and timetable data are securely and efficiently managed in a database.

Version Control: Source code running, changes recording, and teamwork facilitation are achieved through GitHub.

Project Management: Agile Scrum techniques are used to plan sprints, track progress, and do tasks efficiently.

Testing: We have tested the system functionalities to the full extent, so an accurate timetable is generated and a conflict, free timetable is produced.

3.6 Sprint Breakdown

The Smart Classroom and Timetable Schedular's Agile Scrum process was split into several sprints.

In Sprint 1, it was all about gathering requirements, designing the system, and laying out the project plan.

During Sprint 2, the team built the data input parts and merged the database.

In Sprint 3, the focus was on developing the reasoning, based scheduling logic and timetable creation.

Sprint 4 was about building the UI, testing, and polishing it.

Sprint 5 concentrated on checking the work quality, bug fixing, and making the last changes.

Each sprint ended with testing and a review that facilitated continuous improvement and incremental delivery of features.

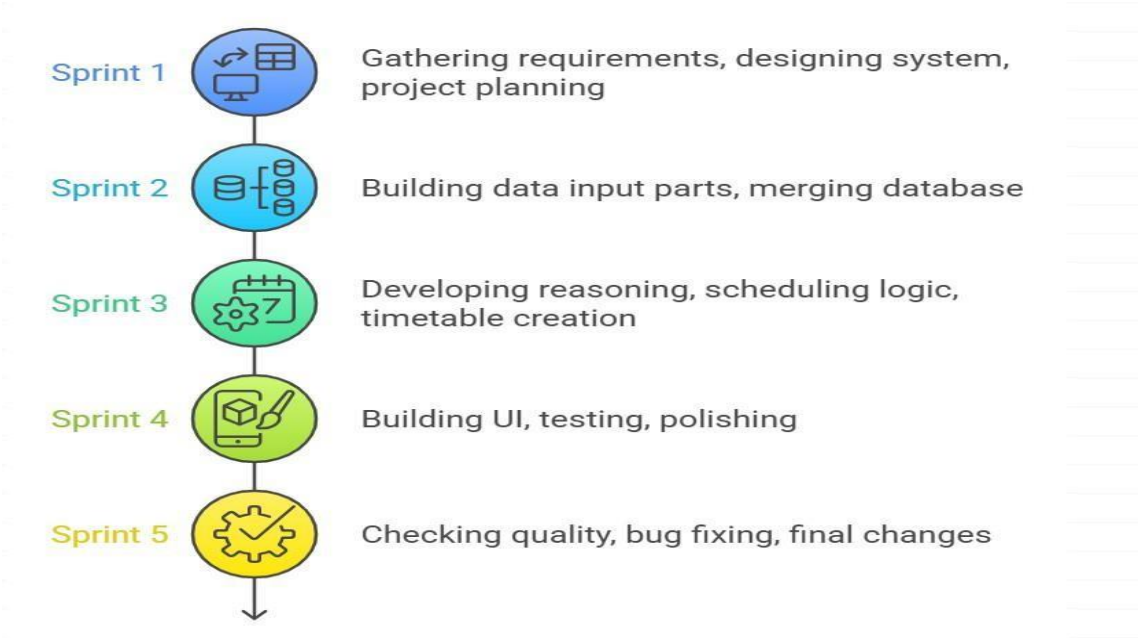

Figure 3.2: Agile Scrum Journey

Chapter 4

Project management

4.1 Project Timeline

The Smart Classroom and Timetable Schedular project timeline is carefully put together to be a structured and well, organized framework that helps ensure the development of the system to be successful. Appropriate timeline planning is extremely important as one of the main factors that contribute to the completion of project activities in a systematic way and within the given timeframe. The mature sequencing of tasks, the optimal use of time, and the regular operation of progress tracking are the features of the timeline throughout the development stage.

The forecasted time frame is in line with the iterative and incremental development principle, giving the possibility of different activities such as analysis, design, implementation, and testing to advance in a mutually agreed manner. This kind of development allows for the early discovery of certain challenges and thus the adjustments can be made on time without impacting the overall project flow. A defined timeline allows the project to maintain consistency, control, and harmonious coordination among the development activities.

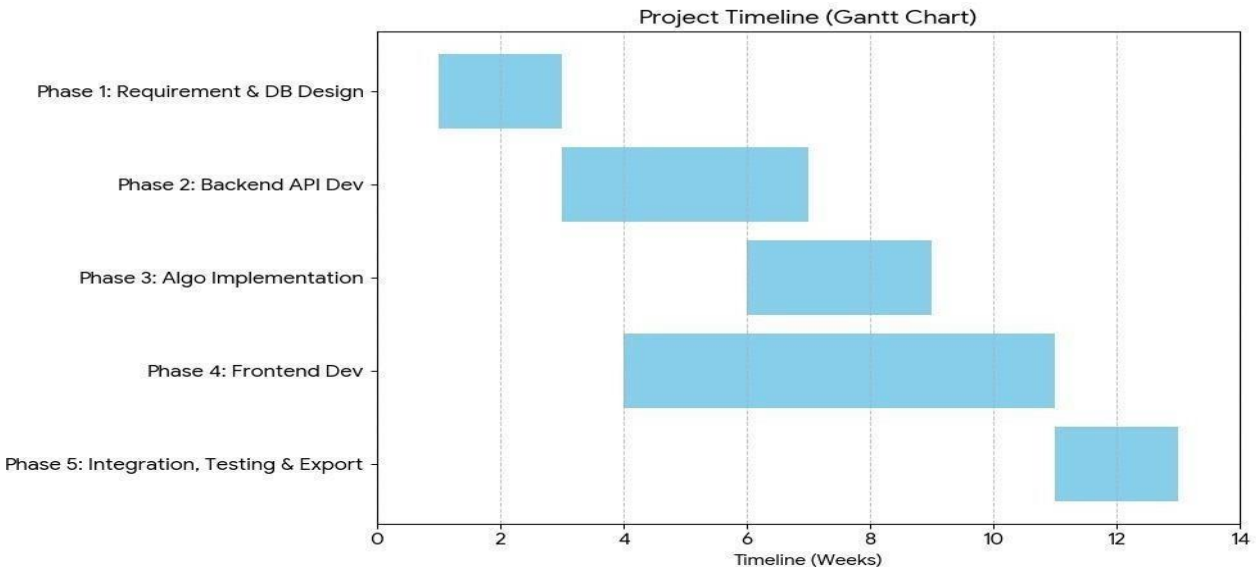

Figure 4.1: Project TimeLine

4.2 Risk Analysis

Risk analysis is the process of identifying potential risks in a project and planning the actions to reduce and control such risks. It is carried out to identify predicted risk factors that could contribute to project failure or delay in which eleven factors were analyzed as major risks. A major risk in the Smart Classroom and Timetable Schedular project case is the use of incorrect or partial data inputs leading to the generation of an inaccurate schedule. A possible way of minimizing the risk is to verify the validity of all inputs before the operations.

Table 4.1: Risk Analysis

Risk ID	Category	Risk Description	Impact	Likelihood	Mitigation Strategy
R1	Technical	Integration issues between backend scheduling logic and frontend user interface	High	Medium	Use modular architecture, API-based integration testing, and continuous debugging
R2	Data	Incorrect or incomplete input data leading to inaccurate timetable generation	High	Medium	Validate input data, apply data consistency checks, and enforce constraints
R3	Algorithm	Increased complexity in handling multiple scheduling constraints simultaneously	High	Medium	Optimize constraint-based logic and perform iterative testing
R4	Time Management	Delays in development due to requirement changes or algorithm tuning	High	Medium	Follow Agile Scrum with weekly sprints and regular progress reviews
R5	Performance	Slow timetable generation for large datasets	Medium	Medium	Optimize backend processing and

Risk ID	Category	Risk Description	Impact	Likelihood	Mitigation Strategy
					improve data handling efficiency
R6	User Interface	Poor usability or accessibility for non-technical users	Medium	High	Design a simple, responsive UI and conduct user testing
R7	Deployment	Configuration issues during system deployment	Medium	Low	Test deployment in a controlled environment before final release
R8	Maintenance	Difficulty in scaling the system for larger institutions	Medium	Medium	Use modular and scalable design to support future expansion

4.3 Project Budget

The Smart Classroom and Timetable Schedular project has been structured in a way that it is affordable, mainly relying on open, source tools and technologies. As the system is built using programming languages, databases, and frameworks that are all freely available, the overall cost of development is very low. The cost of the set, up is mainly the price of the hardware that would be needed such as a PC or laptop, internet and electricity.

There are no licensing costs added as a result of the use of open, source software and GitHub for version control. The major input of the project is human work in terms of development and testing time. So, Smart Classroom and Timetable Schedular is a proof, of, concept of a highly efficient and scalable academic scheduling system capable of being developed at a very minimum financial cost, thus making it a perfect candidate for adoption by educational institutions with tight budgets.

Chapter 5
Analysis and Design

5.1 Requirements

The Smart Classroom and Timetable Schedular system requirements are a set of stipulations that characterize the functional and non, functional aspects of the timetable generation process which is very efficient. Functionally, the system must give the administrators the ability to enter the faculty details, subjects information, classroom facilities, and time slots. It should be generating conflict, free timetables automatically while ensuring proper allocation of resources and faculty workload balancing.

Moreover, the system should enable modification and regeneration of timetables after any changes.

Non, functional requirements are those related to usability, scalability, reliability, and performance of the system. The system should be user friendly for non, technical users, capable of handling increasing amounts of data, and able to generate timetables efficiently without delays. Besides that, security and data integrity are very important to provide ways for safe storage and handling of academic information.

5.2 Block Diagram

The Smart Classroom and Timetable Schedular block diagram illustrates the main components of the system and how they work together. The system takes off at the User/Admin Input Module, where it receives academic data.

This data is kept in the Database Module, which serves as the central storage. The Scheduling Engine uses constraint, based logic to sift through the data and come up with a valid timetable. The answer is passed to the User Interface Module, where one can see, manage, and even export the generated timetable.

Such a modular architecture guarantees simplicity, ease of maintenance, and the possibility of extension.

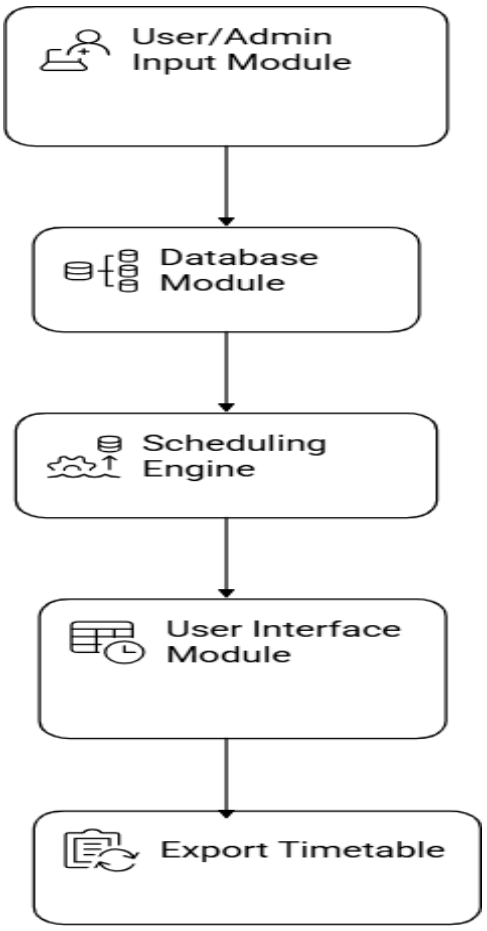

Figure 5.1: Block Diagram

5.3 System Flow Chart

The system flow chart briefly outlines how each step in the Smart Classroom and Timetable Schedular system is carried out. At first, the user is authenticated and then the data is input and validated. After the inputs have been checked, the scheduling engine gets the constraints that have been set and creates the timetable. The system then looks for conflicts and if there are any, the process is repeated until a conflict, free timetable is achieved. At the end, the timetable is shown to the user and saved in the database. The flow here guarantees orderly implementation and gradual fixing of errors.

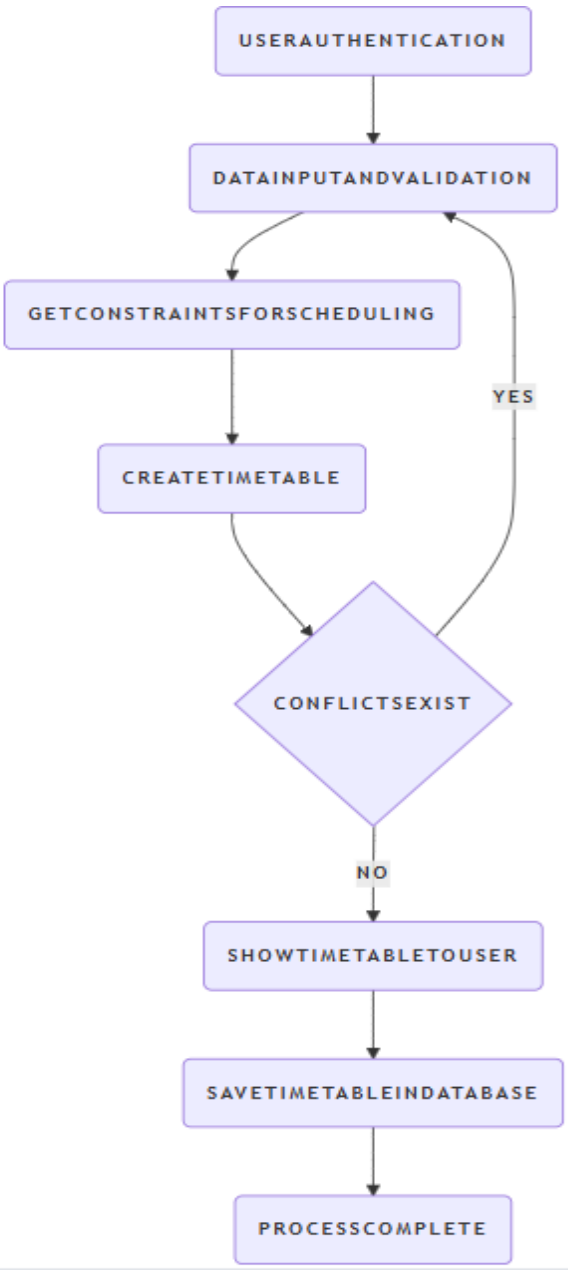

Figure 5.2: System Flow Chart

5.4 Choosing devices

Table 5.1: Choosing Device

Device / Component	**Purpose**	**Suggested Model / Technology**	**Rationale**
User Devices	Client-side access for students, faculty, and admins	Web Browser (Desktop / Mobile)	Platform-independent, no installation required, accessible from anywhere.
Faculty/Admin Devices	Timetable creation, updates, and monitoring	Laptop / Desktop with modern browser	Larger screen improves usability for managing schedules and analytics.
Edge / Local Display (Optional)	On-campus timetable display	Raspberry Pi 4 + HDMI Display	Low cost, energy efficient, suitable for continuous display in corridors or notice boards.
Backend Compute	Application logic & scheduling engine	Cloud Server (Firebase Cloud Functions / Node.js backend)	Scalable, reliable, and supports real-time operations with minimal maintenance.
Database Server	Persistent timetable and user data storage	Firebase Firestore / PostgreSQL	Firestore enables real-time updates; PostgreSQL supports structured and complex queries.
Authentication Service	Secure user login and role management	Firebase Authentication	Simplifies authentication, supports email/password and role-based access.
Notification Service	Alerts for timetable updates	Firebase Cloud Messaging / Email API	Enables instant notifications to users across devices.
QR Generator (Optional)	Timetable sharing and verification	QR Code Library (JavaScript-based)	Easy generation of timetable links and quick access via scanning.
Deployment Platform	Hosting frontend and	Firebase Hosting /	Fast deployment, CDN support,

Device / Component	**Purpose**	**Suggested Model / Technology**	**Rationale**
	backend services	Vercel	and seamless GitHub integration.

5.5 Designing units

5.5.1 User Interface Unit

This unit offers a web, based user interface for students, faculty, and administrators to view and manage the timetables. It guarantees easy to use, responsive, and accessible features over desktop and mobile devices.

5.5.2 Authentication and Authorization Unit

This unit handles secure user login and role, based access control. It prevents unauthorized users from creating, editing, or viewing timetables and only allows among the users, who are permitted and based on their roles.

5.5.3 Timetable Generation Unit

This unit represents the main function of the system. It creates optimized timetables by taking into account various constraints such as the allocation of the subjects, the availability of the faculty, the capacity of the classrooms, and the slots of time.

5.5.4 Database Management Unit

It is a unit that deals with the saving and opening of all the data from the system such as user profiles, schedules, subjects, and constraints. It is the one responsible for data consistency and dependability.

5.5.5 Notification Unit

The notification unit communicates with the users through live messages and updates them of the timetable changes or any other significant information thus facilitating smooth communication.

5.5.6 Analytics and Monitoring Unit

The unit is in charge of tracking the system usage and performance. It gives data that enable the administration to know how scheduling is efficient and how reliable the system is.

5.5.7 Deployment and Maintenance Unit

This unit is in charge of the system hosting, updates, and maintenance.

It guarantees high availability, scalability, and trouble, free system operation.

5.6 Standards

Smart Classroom and Timetable Schedular is a product developed with a commitment to rigorous software engineering standards and principles of quality and reliability. The system was developed in compliance with coding standards and modular architectural principles, as well as following the rules of database normalization. Most importantly, the web application complies with all the standards such as secure data handling, input validation, and responsive design. By using open, freely available and generally accepted standards, Smart Classroom and Timetable Schedular is a product that guarantees compatibility, maintainability, and long, term usability.

5.7 Mapping with IoT WF reference model layers

5.7.1 Physical Devices and Controllers Layer

The layer comprises devices used by the users like laptops, desktops, and mobile phones. Besides, on, campus display units like screens based on Raspberry Pi can also be a part of it.

5.7.2 Connectivity Layer

The connectivity layer is responsible for the transmission of data via the internet and Wi, Fi networks, thus communication is always possible through safe HTTPS protocols.

5.7.3 Edge Computing Layer

Some basic operations like input validation and temporary data caching are done at the client side in order to lighten the server load and enhance the user response time.

5.7.4 Data Accumulation Layer

This layer comprises persistent databases that store timetable data, user information, and frequently scheduling constraints such as Firestore or PostgreSQL.

5.7.5 Data Abstraction Layer

It's the layer where back, end services handle and prepare the data they have gotten. Different timetable creation algorithms here ensure the application of various scheduling rules and constraints.

5.7.6 Application Layer

It is at this layer that the users can freely run the web application with whom they can create, see and manage timetables.

5.7.7 Collaboration and Processes Layer

This layer is designed to facilitate and streamline the academic workflows such as timetable approvals, notifications, and communications between students, faculty, and the administration.

5.8 Domain model specification

The domain model is essentially a conceptual framework of Smart Classroom and Timetable Schedular Automated Timetable Generation System, which identifies the main entities of the system and defines the relationships between them. It illustrates at a glance how the real, world concepts of timetable scheduling have been translated into the software components. The domain model is very helpful in comprehending the system requirements and it is also the basis for the database design and system implementation.

With Smart Classroom and Timetable Schedular, the domain model delineates academic scheduling components such as users, faculty members, subjects, departments, time slots, classrooms, and timetables. These entities stand for the fundamental concepts of an academic environment and they interact with one another in order to facilitate automated timetable generation.

Department: The Department object models academic departments of the institution. Each department is related to multiple subjects, faculty members, and timetables. This entity is used to structure academic data and to implement department, specific scheduling rules.

Subject: The Subject entity is a record of courses offered by various departments. Some of the characteristics are subject code, subject name, department, faculty assigned, and number of sessions required. The timetable generator assigns subjects to suitable time slots accordingly.

Time Slot: The Time Slot is an entity that illustrates the working days and hours which are available to be scheduled. Days, start and end times, and the availability status are a few of the attributes. Time slots are a great tool for ensuring that subjects are evenly distributed across the academic week.

Classroom: The classroom entity is a representation of the real or online teaching spaces. Some attributes are room number, capacity, and availability. This entity acts as a check to make sure that no classroom is given several sessions simultaneously.

Timetable: The Timetable entity is a depiction of the final schedule which is generated. It keeps records of the relationships between different subjects, faculty, classrooms, and time slots. Each timetable line is a guarantee that criteria are met and conflicts are prevented.

Relationship Between Entities

- A department can manage several subjects and faculty members.
- Faculty might teach multiple subjects, but they cant be scheduled at the same time.
- Each subject belongs to a single department and can be assigned to one or more time slots.

Purpose of the Domain Model

The domain model helps to clarify system design in detail by defining the system boundaries and showing the system responsibilities. Developers get a clear view of entity interactions and therefore, it is helpful in maintaining consistency for requirements, database schemas, and

application logic. Having a well, crafted domain model, Smart Classroom and Timetable Schedular could create, scale, and generate accurate schedules with less trouble.

5.9 Communication Model

The communication model illustrates how the elements of the Smart Classroom and Timetable Schedular automated timetable generation system work together to share data and coordinate system operations. It neatly lays out the channels of information exchange between users, system interfaces, backend processing units, and the database. This model is key to grasping the system's behavior at runtime and the cooperation of different modules for the automated timetable generation.

Communication in Smart Classroom and Timetable Schedular is based on the client server architecture, with the client side focusing on user interaction and the server side on processing and data storage. The separation of tasks like this is a sure way to get better system organization, heightened security, and simpler maintenance.

Actors involved in Smart Classroom and Timetable Schedular communication model consists of major actors and elements, which are:

Administrator/User: The administrator is the main user of the system. He/ She is the one who carries the academic data management and for this running the system. The administrator can interact with the system to enter faculty details, define subjects, specify constraints, generate timetables, and view results. The administrator's communication initiations lead to the starting point of the majority of operations in the system.

Web Browser (Client Interface): The browser acts as the client interface with which the users engage Smart Classroom and Timetable Schedular. It shows forms, dashboards, and timetable outputs. The browser sends user requests to the server and receives responses as web pages or messages.

Web Interface Layer (Frontend): The frontend layer that is made with HTML, CSS, JavaScript, and JSP, is the user and backend communication mediator. It gathers user inputs, makes a few basic checks, and sends the requests to the application server.

Application Server (Backend): The backend server handles processing requests that come from

the outside. It holds the business logic for the generation of timetables, constraint validation, session management, and communication with the database. This module is the hub of the communication model.

Database Server: The database server is where all the data that needs to be saved continuously for the system is kept, such as user credentials, faculty information, subjects, time slots, classrooms, and timetables that have been generated. It answers the application server's queries.

Overall Communication Architecture:

Smart Classroom and Timetable Schedular 's communication architecture is based on a request response model. Basically, when a user takes an action, a request is generated and passed along the frontend to the backend for processing. Then the processed data is returned as a response. This design helps in maintaining clear paths of communication and also minimizes the complexity of the system.

The breakdown of the interaction into three layers is a very general way of describing it:

User Layer: Administrator or authorized user

Application Layer: Web interface and backend processing

Data Layer: Database server Each layer communicates only with the surrounding layers.

This way, the system components remain independent and the data flow is always controlled.

Stepwise the communication flow in Smart Classroom and Timetable Schedular system can be grasped from the following points:

A user initiates contact with the Smart Classroom and Timetable Schedular system by entering the URL of the system in a web browser.

The login page is shown to a user by the web interface, and it takes the user's credential input.

The user submits login credentials and sends them to the backend server via an HTTP request.

The backend server records in the database whether the credentials are valid or not.

If the credentials get verification successfully, then the server returns a response enabling the user to access the system functionalities.

The user feeds academic data e.g. faculty details, courses, and time slots through web forms.

The frontend conveys that information to the backend server for processing and storage.

The backend server evaluates the correctness of the submitted data and stores the data in the database.

The user requests the generation of the timetable and this request is first forwarded to the backend server.

The backend gets the timetable generation and checking of the constraints done.

5.10 IoT Deployment Level

The Internet of Things (IoT) deployment level indicates the extent to which a system can be integrated in an IoT, enabled environment architecture comprising data collection, processing, communication, and application layers. Even though Smart Classroom and Timetable Schedular Automated Timetable Generation System is mainly a web, based application, it can be logically aligned with an IoT deployment model to facilitate potential integration with smart campus environments.

Smart Classroom and Timetable Schedular belongs to cloud, centric IoT deployment level, where data is processed and stored centrally, and users connect with the system via network, enabled devices. The system is not a direct physical sensor controller; it is at an application and service level that can utilize data produced by IoT devices, for example, if this is the case in the future.

Deployment Level Description

Smart Classroom and Timetable Schedular today is running at the Application and Cloud Processing Levels in an IoT architecture.

This level of IoT deployment:

Logical IoT Mapping of Smart Classroom and Timetable Schedular

The IoT left layers, for example, can be one system for the Smart Classroom and Timetable Schedular project:

Device Layer (User Devices): User devices like computers and smartphones typically serve as endpoints. These devices are connected to the internet and enable users to access the system via web browsers.

Communication Layer: Communication between users and the Smart Classroom and Timetable Schedular system happens through standard network protocols such as HTTP/HTTPS.

This layer makes sure that data transmission is reliable and secure.

Data Processing Layer (Cloud/Server): The backend server executes timetable generation algorithms, checks constraints, and performs database operations. This layer stands for the brain of the system.

Application Layer: The application layer is responsible for displaying to the user generated timetables, reports, and notifications, etc. This layer facilitates decision, making and the administration of control.

Data is gathered manually or through can be user interfaces rather than sensors.All the processing and decision, taking logics are done on a central server or cloud platform.Users can access the system by using an internet, enabled desktop, laptop, or a tablet.This level is ideal for institutional systems where it is more important to have central data management and real, time access than direct device control.

Current Deployment Level:

Smart Classroom and Timetable Schedular is presently at a level corresponding to an IoT Level where cloud, based data processing and application services are the main focus, without direct integration of sensors. Such a level suits academic scheduling systems since it focuses on data accuracy, centralized control, and accessibility.

Future IoT Integration Possibilities

The system is built with extensibility in mind and can be raised to support higher IoT deployment levels. Some of the future potential integrations might be:

Smart classroom sensors to keep track of the number of people in the roomFaculty or student attendance can be recorded through RFID or biometric systemsAutomatically generated message regarding timetable changes on the smart notice boardsSynchronization in real, time with smart campus management systems. On such occasions, Smart Classroom and Timetable Schedular would get the data from IoT devices and change its scheduling decisions dynamically, thus going towards a more advanced IoT, enabled deployment level.

5.11 Functional view

The functional perspective of the Smart Classroom and Timetable Schedular Automated Timetable Generation System elaborates on the system with respect to its main functionalities and operations. It is centered on the system's actions to facilitate the automated timetable generation, and not on the technical aspects of the system's implementation.

5.11.1 User Authentication and Authorization

This function is very important for providing protected access to the system. Only users who are authorized e.g., the administrators and academic staffs, can log in with their valid credentials. The system will verify the username and password before granting access.

Role, based authorization thus limits the users who can undertake sensitive operations like timetable generation and data modification only to those who have been given the permission. Hence this function is basically a two, way data security measure, i.e., institutional data is protected as well as users' data.

5.11.2 Academic Data Management

An educational data management module equips a school administrator with capabilities to enter, refresh and monitor the school's pertinent academic data.

The data include teachers, courses, divisions, schoolrooms, and schedule slots. Conditions for

validation do not allow users to leave obligatory fields blank or to submit incorrect data formats.

Proper data management results in data integrity and correctness which are the essential factors for coming up with dependable timetables.

5.11.3 Constraint Definition and Handling

This function supports the setting up and the handling of the institutional constraints. Constraints refer to aspects such as faculty availability, the need for a certain number of subject sessions, working hours, and classroom capacity. These constraints are kept in the system and referred to when generating the timetable so that the policies of the institution are followed. If constraints are well dealt with, the number of scheduling conflicts will be drastically reduced, and the resources will be fairly distributed.

5.11.4 Timetable Generation

Timetable generation forms the backbone of Smart Classroom and Timetable Schedular. By applying a rule, based scheduling algorithm internally, the software is able to assign subjects, teachers, classrooms, and time allotments and observe all the set constraints at the same time.

The system makes sure that neither a faculty member nor a classroom gets an assignment more than once during the same time slot.

Such a feature greatly cuts down the amount of labor being done by hand and accelerates the process of making a timetable.

5.11.5 Conflict Detection and Validation

Once the timetable is produced, the system carries out a series of validations to identify potential conflicts.

Basically, this function compares with each other, the teacher's schedule, the usage of classrooms, and the allocation of time slots to keep the situation free from overlap and violation of any kind. Upon identifying the conflicts, the system locks out the finalization step thus errors need to be cleared first.

Consequently, the outcome schedule is not only a correct one, but it is also one that can be put into practice in the real scenario without any issues.

5.11.6 Timetable Viewing and Reporting

This feature enables users to see the generated schedules in an easily understandable and organized manner. Through the web interface, users can get the timetables based on departments or faculties. Good presentation makes it easier to understand and therefore helps users to plan their academic activities. This feature improves the user, friendliness and accessibility of the system.

5.11.7 Error Handling and System Feedback

This function catches cases various and invalid inputs as well as unplanned behavior of the system. Users receive error messages in such cases that help them to understand their mistakes and how they can fix them. The system does not go down, but essentially keeps on operating normally if invalid inputs are given. This function is very significant in the overall dependability of the system as well as the users' happiness.

Table 5.2 Functional View of Smart Classroom and Timetable Schedular

Functional ID	Function Name	Description
5.11.1	User Authentication and Authorization	Validates user credentials and provides role-based access control to ensure secure usage of the system.
5.11.2	Academic Data Management	Manages faculty, subjects, departments, classrooms, and time slot data with proper validation.
5.11.3	Constraint Definition and Handling	Defines and applies institutional constraints such as faculty availability and fixed time slots during scheduling.
5.11.4	Timetable Generation	Generates conflict-free timetables using rule-based scheduling logic.
5.11.5	Conflict Detection and Validation	Checks for faculty clashes, overlapping classes, and room conflicts before finalizing the timetable.
5.11.6	Timetable Viewing and Reporting	Displays generated timetables in a structured, readable format for administrators and faculty.

5.11.7	Error Handling and System Feedback	Handles invalid inputs gracefully and provides meaningful feedback without system failure.

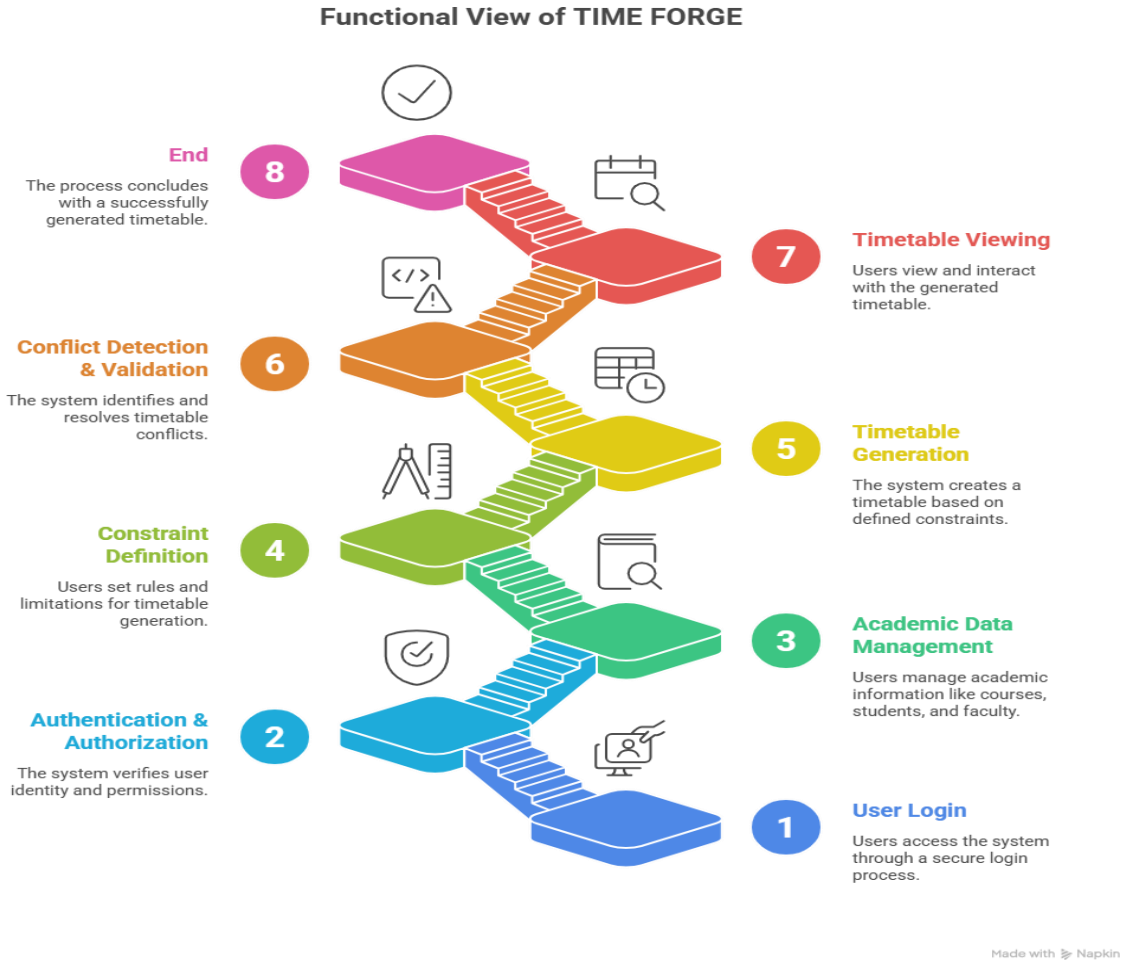

Figure 5.3 Functional View of Smart Classroom and Timetable Schedular

5.12 Mapping IoT deployment level with functional view

The matching of the IoT deployment level with the functional view provides an explanation of how the functional components of the Smart Classroom and Timetable Schedular Automated Timetable Generation System correspond to different layers of an IoT, based architecture. Smart Classroom and Timetable Schedular, being mainly a web, based system, can still be conceptually aligned with an IoT deployment model to illustrate its compatibility with smart campus environments and future IoT integration.

For Smart Classroom and Timetable Schedular the IoT deployment level chosen is the cloud, centric, application, level deployment, at which point data is centrally processed and accessed via network, enabled devices. This deployment level emphasis is on data processing, decision, making, and service provision rather than direct sensor interaction.

Mapping of IoT Layers with Functional Components

User Device Layer (IoT Endpoint Layer)

In the IoT framework, user devices like desktops, laptops, and tablets serve as the IoT endpoints. These are the devices that administrators and faculty members utilize to access Smart Classroom and Timetable Schedular via a web browser.

This layer is a direct implementation of the User Authentication and Authorization (5.11.1) and Timetable Viewing and Reporting (5.11.6) features. Internet, connected devices enable users to interact with the system through logging in, entering data, and viewing generated timetables.

Communication Layer (Network Layer)

The communication layer facilitates the exchange of data between user devices and a central system. Smart Classroom and Timetable Scheduler's communication is achieved through standard web protocols such as HTTP/HTTPS over the internet or a local area network (LAN).

The communication layer indirectly supports the functionalities by allowing secure and reliable transmission of data between the client interface and the backend server. No such functions as data entry, timetable generation, and reporting would have been possible without this layer.

Data Processing Layer (Cloud / Server Layer)

The data transformation layer is the focal point of the system. In Smart Classroom and Timetable Schedular, this layer refers to various core modules like Academic Data Management, Constraint Definition and Handling, Timetable Generation, and Conflict Detection and Validation. All rule, based scheduling logic, constraint evaluation, and validation processes are performed here. The layer checks the generation of the timetable for accuracy, complaint with the institutional policies, and absence of conflicts.

Data Storage Layer (Database Layer)

The database layer is about storing persistent data of the system such as staff data, courses, time slots, constraints, and generated timetables. These two layers; Academic Data Management and Error Handling and System Feedback (5.11.7) are essentially operated through this channel by ensuring data consistency, integrity, and recoverability.

From an IoT perspective, this layer may be regarded as the ultimate single source of truth that underpins analytics and decision, making.

Application Layer (Service Layer)

The application layer is where the services and outputs are given to the user. In Smart Classroom and Timetable Schedular, this layer relates to Timetable Viewing and Reporting as well as general system feedback mechanisms. The application layer changes raw data into valuable information which users can quickly recognize and use.

Significance of the Mapping

The illustration of the IoT deployment line to the functional view proves that Smart Classroom and Timetable Schedular adheres to a layered and scalable architecture. The present system is not making use of physical IoT sensors directly, however, its design is able to accommodate future smart classroom integration, real, time occupancy tracking, or automated notifications through IoT, enabled devices.

This map also shows that each functional component of Smart Classroom and Timetable Schedular can come up logically with a modern IoT, based system model, which guarantees interoperability, scalability, and readiness for smart campus adoption.

Table 5.3 Mapping IoT Deployment Level with Functional View

IoT Deployment Layer	Description	Corresponding Functional View Components
User Device Layer (IoT Endpoints)	End-user devices such as desktops, laptops, and tablets used to access the system	User Authentication and Authorization (5.11.1), Timetable Viewing and Reporting (5.11.6)
Communication Layer (Network)	Enables data transfer between user devices and server using HTTP/HTTPS or LAN	Supports all functional modules by enabling secure client–server communication
Data Processing Layer (Server / Cloud)	Executes business logic and scheduling algorithms	Academic Data Management (5.11.2), Constraint Definition and Handling (5.11.3), Timetable Generation (5.11.4), Conflict Detection and Validation (5.11.5)
Data Storage Layer (Database)	Stores persistent academic and timetable data	Academic Data Management (5.11.2), Error Handling and System Feedback (5.11.7)
Application Layer (Service Layer)	Presents processed data to users in usable form	Timetable Viewing and Reporting (5.11.6), System Feedback

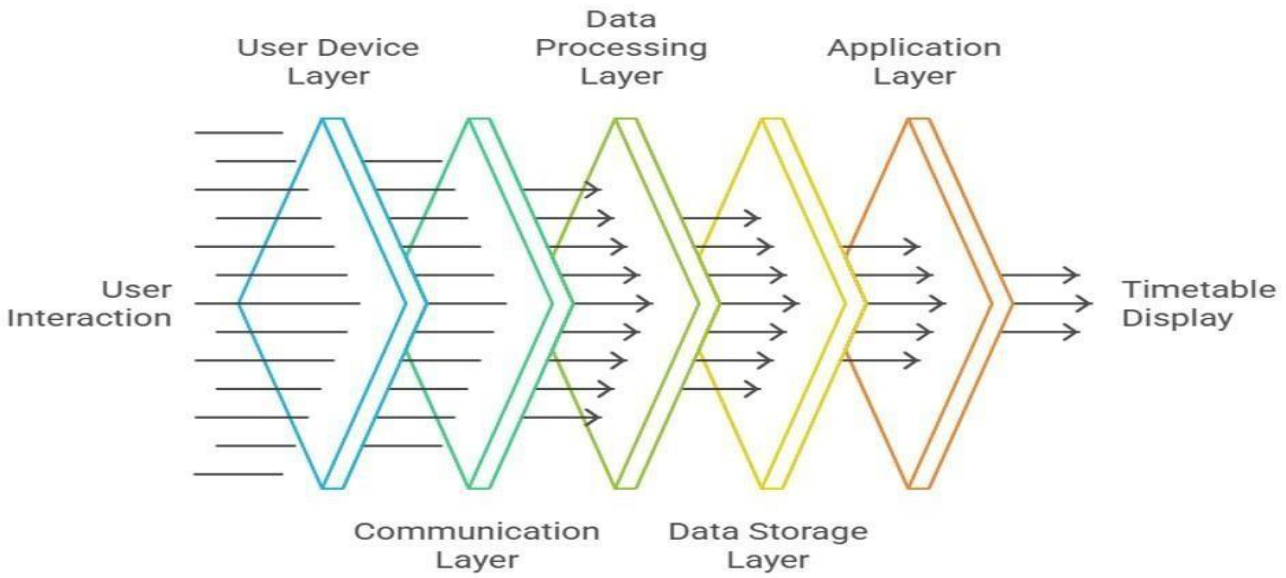

Figure 5.4 Mapping of IoT Deployment Level with Functional View

5.13 Operational view

The operational perspective of the Smart Classroom and Timetable Schedular automated timetable generation system shows the system's features when being used in the actual, real, world setting. The system behavior is emphasized at somewhat regular intervals especially by day, by discussing the interaction of users with the system, the communication of data between different parts, and the system's response to various scenarios. The viewpoint here is more concerned with the concrete operation of the system rather than the abstract design.

System Startup and User Access

The whole scenario begins when the Smart Classroom and Timetable Schedular system is installed on a server and opened to users through a web interface. System administrators and authorized staff members can access the system through regular web browsers over a secure network connection. Upon gaining access to the system, the users will have to authenticate themselves by submitting the correct login credentials.

After verifying their identity, administrators can carry out operations like keying, in faculty, subjects, departments, rooms, and time slots. In addition, institutional constraints like faculty availability and fixed working hours are set at this stage. These operations are conducted via structured forms, and validation checks guarantee the correctness of data before being saved in the database.

Timetable Generation Operation

Upon setting up the data, the administrator triggers the process of timetable generation. The system gets the necessary academic and constraint data from the database and uses rule, based scheduling logic to work out the timetable. Throughout this process, the system verifies the availability of the faculty, the usage of the rooms, and the allocation of the time slots in order to avoid conflicts. If a problem is found, the system will not allow you finalize until the issues have been resolved.

Timetable Review and Approval

After a timetable without conflicts is produced, it is shown to the administrator for a check, up.

Administrators can choose to approve, regenerate, or change the timetable if necessary. It is a way of giving an extra layer of control to the administrators and injecting some flexibility towards unusual academic situations.

Timetable Publication and Usage

After the timetable is approved, it is distributed to the faculty members and other users authorized through the system interface. Faculty members can check their schedules and if changes are made, they are automatically updated in real, time. This promotes unity and proper communication institution, wide.

Error Handling and System Monitoring

The system performs continuous monitoring of user activities and system responses while it is running. The problems related to invalid inputs or unexpected user actions are firstly prevented and then, if they do occur, the system returns to a normal state providing the user with descriptive error messages. System logging and database backups allow for recovery and thus providing continuous service.

Operational Benefits

From an operational perspective, Smart Classroom and Timetable Schedular can be shown as a tool which minimizes manual labor, facilitates more accurate scheduling and through these, ensures the smooth running of academic operations. The system is designed in a way to support growth, thus making it possible for educational institutions to accommodate an increase in data and users without having to undergo drastic alterations.

Table 5.4 Operational View of Smart Classroom and Timetable Schedular

Operational Stage	Description
System Access	Users access the Smart Classroom and Timetable Schedular system through a web browser using a secure network connection.
User Authentication	Administrators and authorized staff log in using valid credentials to gain system access.
Data Configuration	Academic data such as faculty details, subjects, classrooms, and time slots are entered and validated.
Constraint Setup	Institutional constraints like faculty availability and fixed working hours are defined.
Timetable Generation	The system applies rule-based scheduling logic to generate a conflict-free timetable.
Timetable Review	Administrators review the generated timetable and make modifications if required.
Timetable Publication	Approved timetables are made available to faculty and users through the system interface.
Error Handling & Monitoring	System handles invalid inputs gracefully and maintains logs and backups for reliability.

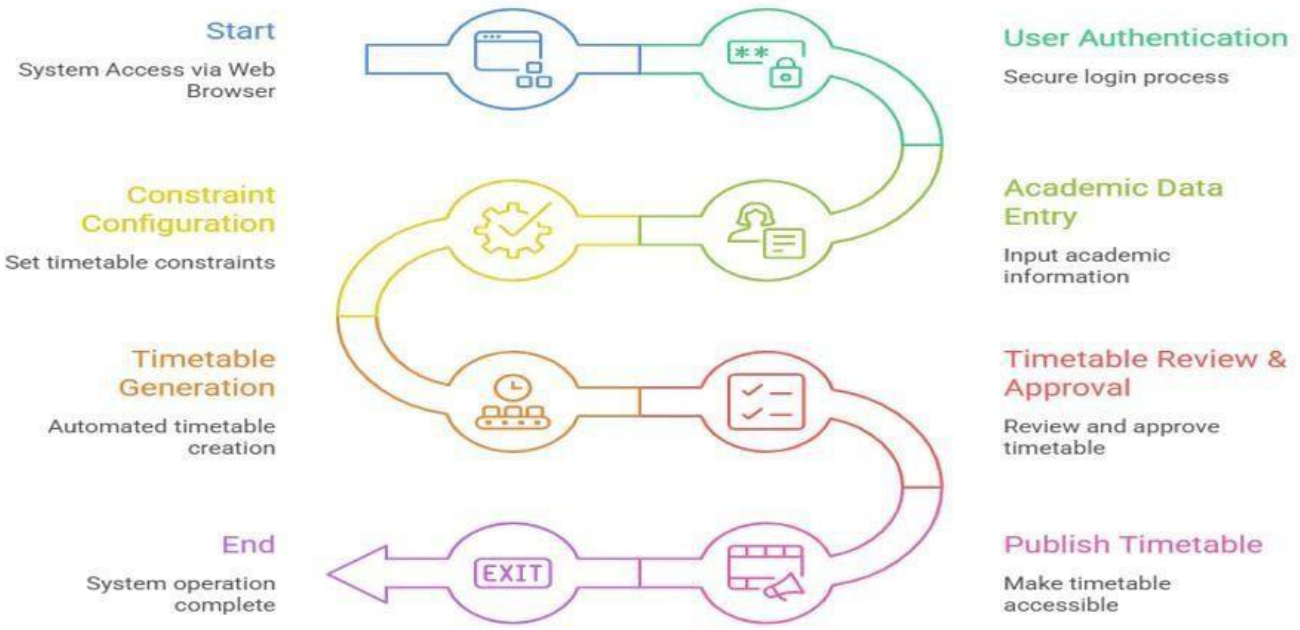

Figure 5.5 Operational View of Smart Classroom and Timetable Schedular

5.14 Other Design

The other design view of the Smart Classroom and Timetable Schedular automated timetable generation system depicts the additional different design factors that might not be purely functional, operational, or related to IoT, views but which, however, play a significant role in the system's robustness, usability, maintainability, and extensibility. All these design elements together elevate the overall quality and performance of the system.

User Interface Design

The user interface of Smart Classroom and Timetable Schedular is basically centered on a very simple and user, friendly design. Since the system is aimed at school administrators and faculty members who may/may not be technically skilled, hence, the interface is kept clean and intuitive.

The forms are easily identifiable, navigation is simple, and the main functions like generating and viewing timetables are seamlessly accessible to the user. The consistent use of design elements throughout the pages helps users to grasp the flow of the application and hence, facilitates the learning process.

Modular Design

Smart Classroom and Timetable Schedular is based on the principle of modular design where different modules like authentication, data management, constraint handling, timetable generation, and reporting split the system. Each module is tailored for a single purpose and communicates with other modules through well, defined interfaces.

Such a layout facilitates system maintenance, makes the process of debugging more straightforward, and permits.

Database Design Considerations

The database design is focused on normalization and data integrity. Different tables are used for users, faculty, subjects, time slots, classrooms, and timetables. Primary and foreign key constraints help to maintain consistency between related data. This design allows for efficient retrieval of data and reduction of unnecessary duplication, which is very important for correct timetable generation.

Security Design

Security is a major design consideration in Smart Classroom and Timetable Schedular. User validation, role, based access control as well as limited database access are the means of securing sensitive academic information. The system design limits performing unauthorized operations and thus it helps in preventing data misuse or accidental modification.

Scalability and Extensibility Design

Smart Classroom and Timetable Schedular has been designed to be scalable, so it would be able to support bigger institutions in the future. The modular design of the architecture enables the system to increase the number of users, departments, and scheduling constraints that it can handle, with minimum changes. Features such as advanced optimization algorithms or smart campus integration may be added as extensions.

Maintainability Design

Using code convention, writing documentation in a clear way, and the great structure of the modules are the biggest contributors to system maintainability. Such a design guarantees that future developers will have no trouble comprehending, modifying, and even extending the system. Relying on well, known technologies also enhances maintenance over time.

Table 5.5 Other Design Considerations in Smart Classroom and Timetable Schedular

Design Aspect	Description	Purpose
User Interface Design	Simple, intuitive, and consistent web interface for administrators and faculty	Improves usability and reduces learning effort
Modular Design	Separation of system into independent modules such as authentication, data management, and scheduling	Enhances maintainability and flexibility
Database Design	Normalized relational database with well-defined relationships	Ensures data integrity and efficient retrieval
Security Design	Authentication, authorization, and controlled data access	Protects sensitive institutional data
Scalability Design	Support for adding more users, departments, and constraints	Enables future growth
Maintainability Design	Clear coding standards and documentation	Simplifies future modifications and upgrades

Figure 5.6 Other Design Overview of Smart Classroom and Timetable Schedular

Chapter 6

Hardware, Software and Simulation

6.1 Hardware

The proper development and the triumphant deployment of the Smart Classroom and Timetable Schedular, heavily rely not only on the software technologies involved but also on the availability of appropriate hardware resources. Hardware has a substantial influence on the system's capability in terms of smooth functioning, data processing efficiency, and thus offering a nice user experience. Considering Smart Classroom and Timetable Schedular as a web, based application scheduled for timetables automation, it is obvious that it does not require high, end or specialized hardware, hence it is both commercially viable and affordable for educational institutions.

Smart Classroom and Timetable Schedular has been devised with efficiency in mind so that it can operate smoothly on everyday computing devices that most people have access to. This design philosophy renders Smart Classroom and Timetable Schedular a perfect fit for schools, colleges, and educational organizations that may not be well, equipped in terms of infrastructure, on the other hand, developers can also use their personal computers for the stages of development and testing. As a consequence of the decision to not depend on costly hardware, the system is still very much accessible and easy to get along with.

In general, the hardware necessities of Smart Classroom and Timetable Schedular can be figuratively divided into three main groups:

server, side hardware, client, side hardware network infrastructure Each aforementioned part has a very significant and indispensable role to play in the overall operation of the system, both in terms of dependability and smoothness.

6.1.1 Server, Side Hardware Requirements

Processor Requirements

Processor is one of the most essential parts of the server, as it is the one that performs all the computations. An Intel Core i3 processor or something equivalent will do well enough for Smart Classroom and Timetable Schedular. The processor is responsible for timetable generation algorithms, database queries, authentication requests, and overall system functioning.

The main scheduling principle in Smart Classroom and Timetable Schedular is based on rules rather than on heavy computations. It is mainly about checking whether the constraints such as the availability of the faculty, the distribution of subjects, and the allocation of time slots are met. Hence the, level system does not need processors that are geared to performance computing. A mid, range processor can comfortably handle the workload even if there are several users accessing the system at the same time.

This kind of processor allows for good performance that is balanced with low hardware costs and as such, Smart Classroom and Timetable Schedular becomes accessible to small and medium, sized educational institutions.

Memory (RAM) and Storage Requirements

Memory (RAM) Requirements Random Access Memory (RAM) is a crucial component that determines the amount of multitasking the server can handle at a time. A minimum 8 GB RAM is advisable for smooth running without any disturbance. Having enough RAM means the server can easily accommodate active user sessions, retrieve database queries faster, and carry out the timetable generation logic without the occurrence of delays. If it is a situation where multiple users are accessing the system simultaneously, a good amount of memory will keep the system performance from getting worse and at the same time, it will give you faster response times. If you are talking about a user count of more than one or two i.e. large institutions with several different departments etc, then you might consider upgrading the RAM even more. Whereas, for regular day, to, day use, 8 GB RAM is more than enough to deliver a smooth running and reliable operating environment. Storage Requirements Storage is basically

the place where your application files, database records, system logs, configuration files, and backups are kept. Smart Classroom and Timetable Schedular needs at least 256 GB of storage space, which can be either HDD or SSD. Although a hard disk drive (HDD) storage solution is a viable option, a solid, state drive (SSD) storage is highly recommended due to its quick data read and write capabilities. SSDs help to lower the time taken for application launching, accessing the database, and the general system performance. This mode of operation is especially significant when it comes to timetable generation or fetching the large data records. Besides, the storage size is also enough to support the scheduled backups, thus ensuring not only the timetable data but the entire system can be safely restored in case of any data loss.

S Smart Classroom and Timetable Schedular is not limited to any single platform as it can function equally on Windows, Linux, or macOS. Among these choices, Linux, based operating systems are generally the first choice for production environments because they are stable, have good security features, and the costs of running are lower.

Linux servers provide more granular control of system resources, better uptime, and a strong ecosystem of web, based applications. However, Windows servers are not left out and can be used quite well, especially in situations where the institution is already acquainted with a Windows, based infrastructure.

This kind of versatility lets the institution pick an operating system that suits their current configuration and their technical expertise.

In general, the server, side hardware configuration has been designed with the necessary features to make sure Smart Classroom and Timetable Schedular runs smoothly, is dependable, and does not have performance issues, even when it is processing huge amounts of data related to timetables.

6.1.2 Client-Side Hardware Requirements

Client, side hardware typically means the devices that administrators, faculty members, as well as other authorized users, deploy to access the Smart Classroom and Timetable Schedular system. Since Smart Classroom and Timetable Schedular is an internet, based application, users don't have to install any special software to use it but can simply run it on any standard web

browser.

This approach greatly facilitates system access and diminishes hardware reliance at the client side. Users can reach the system via desktops, laptops, or other kinds of computing devices that are usually present in educational institutions.

Processor Requirements: A client computer with an Intel Core i3 processor or an equivalent one is sufficient. Since the server takes care of the majority of the processing tasks, the client computer is essentially responsible for displaying the user interface and processing basic input.

Therefore, even entry, level processors can deliver a seamless user experience. Memory (RAM) Requirements The program needs a minimum of 4 GB RAM to function. However, 8 GB RAM is recommended for a better performance, especially when users running multiple applications apart from the browser. Having enough RAM not only facilitates the user experience but also ensures faster page loading and higher responsiveness while working with dashboards, reports, and timetables.

6.1.3 Network Requirements

Network connectivity holds great significance for the performance and accessibility of Smart Classroom and Timetable Schedular. Being a browser, based system, the supporting network infrastructure must be reliable to allow for smooth communication between users and the server.

Internet Connectivity

The internet connection has to be stable, particularly if the system is running on a central server or cloud platform. A reliable connection will allow administrators and faculty members to access the system without any interruptions.

Local Area Network (LAN)

In a school setting, Smart Classroom and Timetable Schedular may be installed on a school server and accessed via LAN. This kind of arrangement speeds up the access, lowers the reliance on the external networks, and increases the data security.

Bandwidth Requirements

Smart Classroom and Timetable Schedular is mainly a text, based application exchanging schedules, reports, etc. As a result, the bandwidth required for the system is not that high. The network remains at a low load even when several users are working with the system simultaneously.

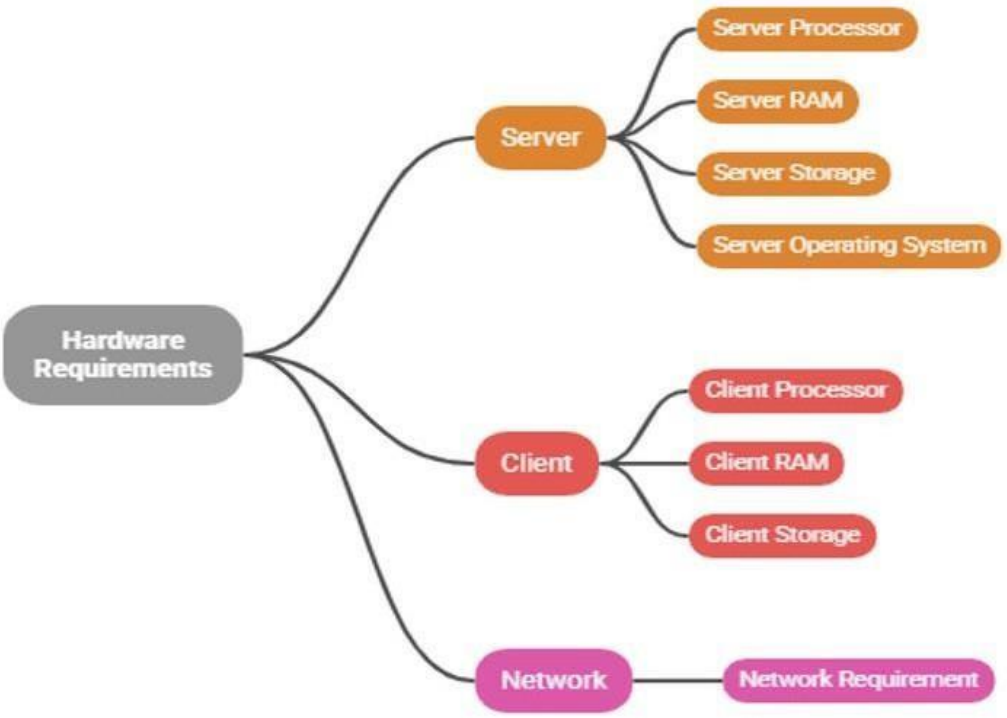

Figure 6.1 *Server-Side Hardware Requirements*

6.2 Software Development Tools

Smart Classroom and Timetable Schedular, a software solution, was developed with the help of reliable and efficient software development tools. These tools were picked very thoroughly so that they can bring about the best of the system while ensuring its future upgrade, friendly nature. Using a variety of tools, especially those popular both in the software industry and academia, gives Smart Classroom and Timetable Schedular a solid and green nature. Programming Language At the heart of Smart Classroom and Timetable Schedular, the main features run in Java. The reason for choosing Java was its excellent platform independence, rich set of object, oriented features, and huge availability of libraries. Besides these, Java helps developers in constructing high, security, scalable, and easily maintainable web applications. Java through its superior performance and stability also plays a very important role in carrying out the scheduling algorithms and other back, end operations.

Web Technologies

Web Technologies Standard web technologies like HTML, CSS, and JavaScript are the tools used for the development of the client, side of a web application. HTML is the fundamental language of a website; it contains all the elements making up a web page. CSS, on the other hand, is the designer that makes everything beautiful and stylish. Lastly, JavaScript is responsible for making the websites lively by permitting interactions, validations, and animations. In conjunction, these three technologies have been used in Smart Classroom and Timetable Schedular to create a user interface that is user, friendly and responsive for the administration and faculty staff.

Backend Framework

Processing on the server side is mostly done through Java Server Pages (JSP) and Servlets. JSP is used for generating web content dynamically, while Servlets perform the tasks of handling requests, managing sessions, and facilitating client, server communication. The combination of these technologies results in a highly responsive and smooth interaction between the user interface and the database.

Database Management System

The system is built on MySQL database management system. MySQL being one of the most widely used relational databases is trustworthy, secure, and performs tasks with great efficiency. It is used for storing data of users, faculties, subjects, timetables, and also for saving system settings. MySQL's query language is so powerful that it makes the data handling operations very easy and efficient.

Integrated Development Environment (IDE)

The development of the application is done in Eclipse IDE. Eclipse is a great environment, offering many features like syntax highlighting, debugging tools, project management support, etc. It simplifies the process of coding and testing such that developers can rapidly find and fix bugs.

Server and Deployment Tools

Apache Tomcat is a web server and servlet container. Tomcat helps to run Java, based web applications and so offers a safe environment for the deployment and operation of the Smart Classroom and Timetable Schedular system. Furthermore, it handles the HTTP requests and ensures that the client and backend components stay continuously communicating without any interruptions.

Version Control System

Git is the version control instrument that has been used to enable the team to handle changes in the source code in the most effective manner. Besides tracking changes, recording code history, and easing collaboration during development, Git also provides a safeguard for the code by offering the option of reverting to the previous version if needed.

Testing Tools

Testing has primarily been manual testing to ensure that the system functions as expected, the generated timetable is correct, and that the whole system goes smoothly without errors. Thus, testing through a web browser is paving the way for assuring that the system is compatible with various platforms and devices.

Figure 6.2 Client-Side Hardware Requirements

6.3 Software code

Java:

```
public class TimetableGenerator {

    public boolean generateTimetable(List<Faculty> faculties,
                                     List<Subject> subjects,
                                     List<TimeSlot> timeSlots,
                                     List<ClassRoom> rooms) {

        for (Subject subject : subjects) {
            for (TimeSlot slot : timeSlots) {

                if (isFacultyAvailable(subject.getFaculty(), slot)
                        && isRoomAvailable(rooms, slot)) {

                    assignSlot(subject, slot);
                    break;
                }
            }
        }
        return true;
    }

    private boolean isFacultyAvailable(Faculty faculty, TimeSlot slot) {
        return !faculty.getAssignedSlots().contains(slot);
    }

    private boolean isRoomAvailable(List<ClassRoom> rooms, TimeSlot slot) {
        for (ClassRoom room : rooms) {
            if (!room.getOccupiedSlots().contains(slot)) {
                return true;
            }
        }
        return false;
```

```
    }
}
```

Servlet / Controller Code (Backend Flow)

```
@WebServlet("/generateTimetable")
public class TimetableServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request,
                          HttpServletResponse response)
                          throws IOException {

        TimetableGenerator generator = new TimetableGenerator();
        boolean status = generator.generateTimetable(
                FacultyDAO.getAll(),
                SubjectDAO.getAll(),
                TimeSlotDAO.getAll(),
                RoomDAO.getAll()
        );

        if (status) {
            response.sendRedirect("viewTimetable.jsp");
        } else {
            response.sendRedirect("error.jsp");
        }
    }
}
```

Database Connectivity Code

```
public class DBConnection {

    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/timeforge",
```

```
            "root",
            "password"
        );
    } catch (Exception e) {
        e.printStackTrace();
    }
    return con;
  }
}
```

JSP Page (Frontend – Viewing Timetable)

```
<table border="1">
  <tr>
    <th>Day</th>
    <th>Time Slot</th>
    <th>Subject</th>
    <th>Faculty</th>
  </tr>
  <c:forEach var="entry" items="${timetable}">
    <tr>
      <td>${entry.day}</td>
      <td>${entry.slot}</td>
      <td>${entry.subject}</td>
      <td>${entry.faculty}</td>
    </tr>
  </c:forEach>
</table>
```

6.4 Simulation

Smart Classroom and Timetable Schedular simulation is critical to verify if the toola works as intended and can produce correct results before it gets implemented in a real educational establishment. Here running the developed system on sample as well as real data is meant to determine how the test schedule generation method performs under various situations and limitations.

The main purpose of test runs in the case of Smart Classroom and Timetable Schedular software is to make sure that the system output reliable, conflict, free class schedules according to the rules of the educational institution. As the timetable allocation process requires dealing with multiple variables like faculty availability, subject distribution, working hours, classroom allocation, etc., simulation helps in forecasting target detection of logical errors, constraint violations, and lower performance points at an early stage.

For simulation, the system is fed with predetermined input data that represent departments, subjects, faculty members, and time slots. The nature of these inputs closely match those of a real institution. The system then utilizes its rule, based scheduling logic to process the data and generates a timetable. The obtained results are thoroughly checked to confirm all constraints are met.

A number of test scenarios were simulated during the series of experiments. Among the scenarios simulated were those with limited faculty availability, a single department having multiple subjects, and overlapping time slots conditions.

Simulation was also instrumental in determining the system's reaction time and overall performance. When the number of simulated users and timetable entries was gradually increased, the system's performance under light stress conditions was noticed. The system was able to work efficiently without any major problems and it took only a reasonable amount of time for it to generate the timetables, thus confirming that it is a good fit for an institution.

Moreover, simulation was employed to corroborate the user experience. The activities of administrators entering data, generating timetables and faculty members viewing timetables were carried out through the website interface. This made it possible to ensure that the presentation layer, business logic layer and database were communicating effectively.

During simulation, if there were any inconsistencies or surprising results, scheduling rules were adjusted and validation checks were enhanced to correct disruptions. This cyclic simulation method has elevated the systems dependability and accuracy to a great extent.

Table 6.1 Simulation Test Scenarios

Scenario ID	Input Condition	Simulation Description	Expected Outcome	Result
S1	Normal Faculty Availability	Timetable generated with sufficient faculty and time slots	Conflict-free timetable	Pass
S2	Limited Faculty Availability	Few faculty available for multiple subjects	Timetable generated without clashes	Pass
S3	Multiple Subjects per Department	High number of subjects for a single department	Balanced subject distribution	Pass
S4	Overlapping Time Slots	Conflicting time slots provided	Conflict detected and resolved	Pass
S5	Multiple Users Access	Simulated concurrent user requests	Stable system performance	Pass
S6	Large Dataset	Increased number of faculty and subjects	Timetable generated within acceptable time	Pass

Chapter 7

Evaluation and Results

7.1 Test Points

The Smart Classroom and Timetable Schedular is a testing phase very important part of the assessment. The main goal of testing is to determine if the system works properly, if it gives accurate results, and if it is in line with all the functional and non, functional requirements, which were laid out in the design phase. The operations of timetable generation, system reliability, usability, and performance under realistic conditions are the main points of the testing process.

A group of well, defined test points was selected to thoroughly assess the system. Each test point is concerned with an individual feature or function of the system. For the system testing, both real and institutional simulated data were utilized to guarantee practical application.

7.1.1 User Authentication Test

This aspect of testing is concerned with determining whether the system has the capability to correctly recognize its users such as administrators and authorized staff members.

Only legitimate users whose authentic credentials are recognized by the systems should be allowed to log in. The system should return an error message whenever the user enters the wrong username and/or password. Session management should be used to ensure that only authenticated users can access the system.

Result:

The results indicated that the system was able to identify the valid users and at the same time, it blocked the invalid one's credential, wise, so the system access was secured.

7.1.2 **Data Entry and Validation Test**

This test checks if the system can properly take and validate input data such as faculty details, subjects, departments, and time slots. Mandatory fields should not allow empty values. Data with incorrect formats should be rejected. Duplicate entries should be properly managed.

Result: The input validations worked perfectly hence no incorrect or incomplete data was stored in the database.

7.1.3 Timetable Generation Test

This is the most crucial test point of the system. It confirms if Smart Classroom and Timetable Schedular produces conflict, free and valid timetables. Faculty members should not be required to teach several classes at the same time. Classrooms should not be booked for two different purposes at the same time. The distribution of the subjects should be done as per the set of rules.

Result: The system was able to produce correct timetables without conflicts and met all the constraints.

7.1.4 Constraint Handling Test

This test is on how the system performs in the presence of various institutional constraints.

Result:

Smart Classroom and Timetable Schedular took great care of all the constraints and no violations happened during timetable generation.

7.1.5 Database Interaction Test

This is the test point that confirms whether the application and the database are communicating correctly. The operations of data insertion, retrieval, update, and deletion should all be working properly. The data consistency should be maintained.

Result:

All database operations were performed successfully with accurate data storage and retrieval.

7.1.6 User Interface Test

The objective of this test is to see how easily the user can operate the system and how quickly the interface reacts.

Different browsers should be able to render the pages correctly. The navigation menu should be clear and easy to follow. The class schedule should be presented in a very simple understandable way. UI was very intuitive, quick in responding, and visually clear.

7.1.7 Performance Test

This test evaluates the performance of the system under the moderate load.

The duration taken to generate a timetable should be a sensible one. The system should be capable of handling multiple users simultaneously without any downtime. The system has operated effectively, and there have been no issues of instability in response times throughout the entire testing period.

7.1.8 Error Handling Test

This test evaluates the system's capability to handle and recover from user mistakes and system failures. The system should display friendly error message to the user. Even when there are invalid inputs, the application should not go crashing. The system was efficient in handling errors and at the same time the overall stability of the system was not compromised.

Table 7.1 Test Points and Results

Test Point ID	Test Name	Description	Expected Outcome	Actual Result	Status
TP1	User Authentication Test	Verification of login using valid and invalid credentials	Only authorized users should access the system	Valid users allowed, invalid users blocked	Pass
TP2	Data Entry and Validation Test	Validation of faculty, subject, department, and time slot data	Invalid or empty data should be rejected	Incorrect and duplicate data prevented	Pass
TP3	Timetable Generation Test	Generation of timetable based on constraints	Conflict-free timetable generation	Timetable generated without conflicts	Pass
TP4	Constraint Handling Test	Handling faculty availability and fixed constraints	Constraints must be strictly followed	All constraints satisfied	Pass
TP5	Database Interaction Test	Testing insert, update, retrieve operations	Accurate and consistent data handling	Data stored and retrieved correctly	Pass
TP6	User Interface Test	Checking usability and navigation	Simple and user-friendly interface	Interface responsive and clear	Pass
TP7	Performance Test	System behavior under moderate load	Stable response time	No performance degradation	Pass

Test Point ID	Test Name	Description	Expected Outcome	Actual Result	Status
TP8	Error Handling Test	Handling invalid inputs and system errors	No system crash, proper messages	Errors handled gracefully	Pass

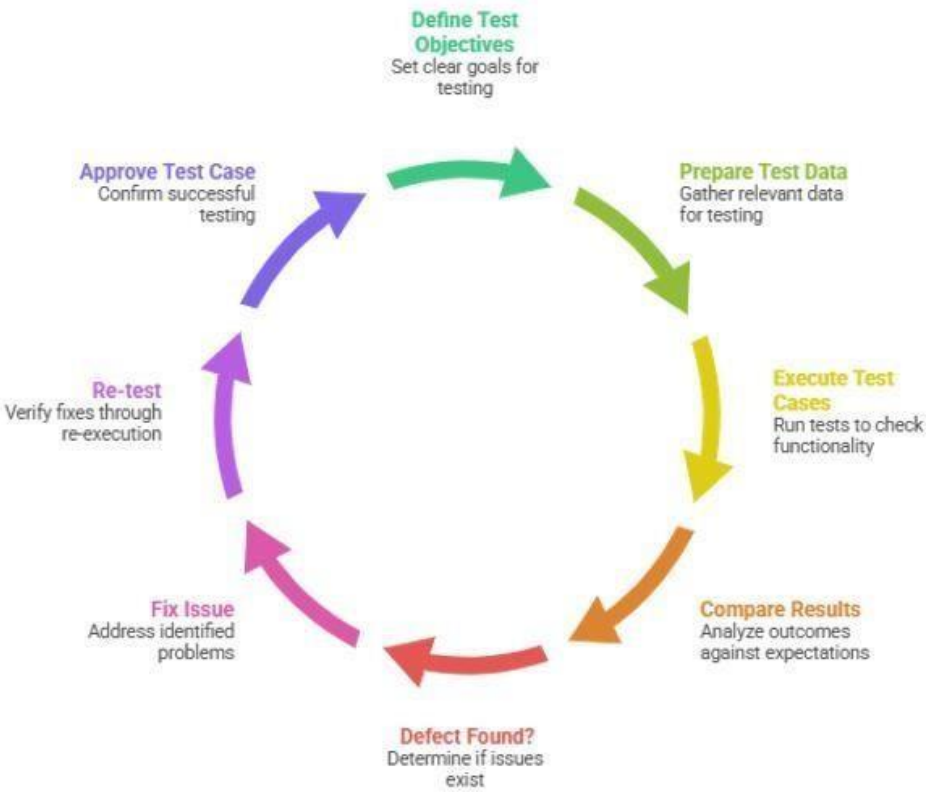

Figure 7.1 Testing Process Flow

7.2 Test Plan

The test plan outlines a methodical approach that has been adopted to test and validate the various features, performance, and dependability aspects of the Smart Classroom and Timetable Schedular Automated Timetable Generation System. This test plan aims to confirm that the system complies with the requirements and operates correctly in the expected working environment prior to its release.

The tests that were scheduled beforehand, and the entire test cycle has been undertaken in a very systematic way, which includes testing of the main modules of the system such as user login, data handling, timetable generation, constraint management, and user interaction with the interface. The tests considered the functional as well as the non, functional attributes of the software.

7.2.1 Test Objectives

The overall objectives of the testing strategy are:

To check that all features of the system are working properly. To test the logic of the timetable generation in order to make sure it is correctly and conflict, free producing the schedules. To observe and verify system performance in both normal and extreme situations. To make sure that the data is consistent and the system interacts correctly with the database. To assess the systems efficiency, ease of use, and performance.

7.2.2 Test Environment

Smart Classroom and Timetable's environment and equipment were emulated as closely as possible to the real ones in order to conduct tests.

Hardware environment:

Ordinary desktop or laptop computers with minimum specification requirements.

Software Environment:

Operating System: Windows / Linux

Web Server: Apache Tomcat

Database: MySQL

Browser: Google Chrome / Mozilla Firefox

This setting provided the testing team with a very close real scenario and it was also possible to get a very accurate system's reaction assessment.

7.2.3 Test Strategy

The test strategy was based on a modular and incremental approach that finally led to overall integration. At first, the testing of each module was done separately, and then the combined testing was carried out. The breakdown of the strategy was:

Unit Testing:

Login, data entry, and timetable generation components had almost all been tested individually in order to be corrected if necessary.

Integration Testing:

Testing the interactions between different modules, for instance, frontend and backend, ensuring that the data is flowing properly.

System Testing:

Testing the entire system to verify if it is capable of producing an output that conveys the whole work process from start to finish.

7.2.4 Types of Testing Performed

Functional Testing:

Functional testing pretty much means to show and prove functionalities that are doing what has been documented.

Verification Testing:

Verification testing is basically doing a double check on the inputs and their correctness as well as the outputs obtained and whether they correspond to the expected ones.

Performance Testing:

Performance testing is one of the testing types that focus on the response time and how the system behaves with a light load.

Usability Testing:

Usability testing is a way of finding out how easy it is for a user to use the system, go through its features, and how understandable the interface is.

Error Handling Testing:

Error handling testing is a kind of test that ensures the system does not freeze or crash when an invalid or unexpected input is given.

7.2.5 Test Data

Test data are a mixture of random and real institutional data. The records comprise:

Faculty details Subjects and departments Time slots Classroom information Constraint rules. By using real data not only the logic of the timetable generation was tested but it also became possible to evaluate it.

Table 7.2 Test Plan Overview

Test Type	Description	Method Used	Outcome
Unit Testing	Individual modules tested separately	Manual testing	Modules functioned correctly
Integration Testing	Interaction between frontend and backend	Manual testing	Smooth data flow observed
System Testing	Complete system tested end-to-end	Manual testing	System met all requirements
Functional Testing	Verification of system features	Test case execution	All features working as expected
Performance Testing	Response time under load	Simulation-based testing	Stable performance achieved
Usability Testing	Ease of use and navigation	User observation	Interface user-friendly
Error Handling Testing	Handling invalid inputs	Boundary testing	Errors handled safely

Figure 7.2 Timetable Evaluation Flow

7.3 Test result

Test results from a review of Smart Classroom and Timetable Schedular showed that the system can meet its functional and non, functional requirements. Only under the defined test environment were all the test case preparations executed, and the system is described as reliable, accurate, and consistent throughout the testing process.

During user authentication testing, the system was successful in distinguishing between valid and invalid users. Authorized users have been able to access the system without any problems while the users who tried to login with incorrect details were appropriately given error messages. Session handling methods efficiently stopped unauthorized users from accessing internal modules, thus, access control was very secure.

After data entry and validation testing, Smart Classroom and Timetable Schedular was found to be capable of verifying all the input fields correctly. Required fields did not allow submissions to be empty, data in the wrong format was not accepted, and duplicate entries were prevented. Therefore, only the correct and consistent data were stored into the database, which is the reason for the correct creation of the timetable.

The experiments to generate timetables gave very good results. The system was able to produce timetables without any conflicts by strictly adhering to along the constraints that were defined.

No faculty has been allocated for more than one session at the same time, classrooms have not been double, booked, and the distribution of subjects has been in accordance with the institutional rules. These results indicate that the rule, based scheduling logic that has been implemented in the system is correct.

Besides, when tested for constraint handling, Smart Classroom and Timetable Schedular demonstrated accurate handling of the different institutional constraints such as limited faculty availability, fixed time slots, and department, specific rules. During schedule creation, the system adhered to all constraints, and no violations were found in the produced output.

Tests for database interaction verified that the communication between the application and the database was always running smoothly. All CRUD operations (Create, Read, Update, Delete) have been done correctly, and data consistency has been maintained even after multiple operations. This guarantees reliable data storage and retrieval during actual usage.

User interface testing revealed that the system is intuitive and easy to use. The application was displayed correctly in several browsers, the navigation was simple, and the timetable presentations were neat and well, organized. Users with little technical knowledge could operate the system efficiently.

7.4 Insights

The appraisal and trial of the Smart Classroom and Timetable Schedular exposed a number of very useful insights that covered the depth, breadth, and practicability of the system. The findings came from the observation of the system's behavior during its simulation, testing, and the analysis of its results by means of real institutional data. Among the major lessons learned from the project is the power of automation to minimize human input drastically.

Human creation of timetables is slow and error, prone work, more so when there are many constraints to be considered, such as faculty availability, subject allocation, and classroom usage. Smart Classroom and Timetable Schedular, in fact, automated this process, thus the time needed to come up with a full timetable was drastically shortened, without compromising on accuracy and consistency. An insight closely related to this is the role of constraint, based logic. During testing it was discovered that the process of outlining and verifying constraints is a critical factor in producing conflict, free timetables. The rule, based approach deployed in

Smart Classroom and Timetable Schedular guaranteed, among other things, the prevention of faculty, session, and room conflicts.

This is a clear indication that a well, implemented rule, based system, can efficiently meet the complex scheduling requirements.

The project also brought out the impact of data accuracy in the performance of the system. It was found that the quality of the generated timetable heavily depended on how correct the input data were during the test. Partially or incorrectly filled faculty or subject information are probable to have an impact on the quality of the output. This revelation underscores the importance of having data validation mechanisms, which were successfully embedded in Smart Classroom and Timetable Schedular to reduce such problems.

From a performance standpoint, the system showed a normal level of stability when only moderate load conditions were considered. Even if several users gained access to the system concurrently, the timetable generation jobs were completed quite swiftly within the established time criteria. This piece of information serves as a confirmation that the selected architecture and hardware requirement specifications adequately cover the needs of small to medium, sized educational establishments.

Through usability testing, the team was able to get user experience and system accessibility as feedback. It was revealed that users who are not technically adept can also easily find their way around the web, based interface. Factors like a well, arranged layout, processes that were straightforward, and timetable displays that were easy to read pushed the users toward a good interaction experience. Therefore, this further solidifies the argument of prioritizing the end users while developing systems, especially academic system environments.

One more vital insight obtained is that the system has flexibility for the future extension of its scale. Although at present, the implementation focuses primarily on institutions with moderate student numbers, the feature of a modular software architecture gives the system an additional capability of being extended easily.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

8.1 Social aspects

Smart Classroom and Timetable Schedular, positively impacts society in many ways, especially in schools. It helps the school staff, teachers, and students with their daily challenges by creating timetables automatically, thus, the school becomes more efficient and well, coordinated.

Reducing the manual workload is one of the biggest social contributions of Smart Classroom and Timetable Schedular. Typically, the preparation of timetables demands substantial hours and effort from the academic staff and is often subject to incessant revisions in order to sort out conflicts. Smart Classroom and Timetable Schedular automates this task and, thus, the administrators are freed to concentrate on other academic and management issues much more effectively and without stress.

From the perspective of the teaching staff, the system leads to fair and transparent processes of schedule allocation. Teacher's presence and limitations are automatically factored into timetable creation thus there is no room for any kind of bias or error. Teachers get an equitable number of working hours and thus, their overlap of assignments is eliminated, which results in higher job satisfaction and work, life balance.

Smart Classroom and Timetable Schedular benefits students by improving education and raising the accessibility of it. A good and clear timetable can help students enormously in planning their academic hours. Students can easily get the timetable online. This reduces the confusion and the last, minute changes that often mess up the learning schedule.

The system also supports inclusive institutional practices. Smart Classroom and Timetable Schedular creates the possibility of scheduling scenarios that are sensitive to the well, diversified needs of the students and staff by allowing the accommodation of different constraints such as departmental rules, faculty availability, etc. This flexibility of the system acts as a catalyst for a better classical academic environment.

Improved interaction is another key social factor. When the timetable is centrally managed, it is less likely that there are misunderstandings among departments, faculty, and students, etc. Changes can be done at any time and, once implemented, reflected throughout the system so that all users will always have consistent and accurate information.

Furthermore, Smart Classroom and Timetable Schedular helps in raising digital literacy and technological usage in educational institutions. By automating the prerequisite usage of standardized systems, the project, on its part, facilitates the institution's and users' readiness for modern, technology, driven workflows, thus a step forward to the digital era.

On the whole, Smart Classroom and Timetable Schedular, through its social dimension, brings about positive changes in academic institutions by increasing efficiency, fairness, accessibility, and coordination. It is a socially advantageous timetable management system that supports a well, organized educational setting and at the same time, lowers the administrative load.

Table 8.1 Social, Legal and Ethical Aspects

Aspect	Key Focus Area	Description	Impact
Social Aspect	Workload Reduction	Automation of timetable creation reduces manual effort of staff	Improved efficiency and reduced stress
Social Aspect	Fairness & Transparency	Rule-based scheduling avoids bias in faculty allocation	Better work-life balance for faculty
Social Aspect	Student Accessibility	Online availability of timetables	Reduced confusion and better planning
Legal Aspect	Data Privacy	Controlled access and authentication mechanisms	Protection of sensitive institutional data
Legal Aspect	Intellectual Property	Use of original code and open-source tools	Legal compliance and ownership clarity
Ethical Aspect	Bias Reduction	Automated, constraint-based scheduling	Objective and fair decision-making
Ethical Aspect	Accountability	Human oversight retained in timetable approval	Responsible use of automation

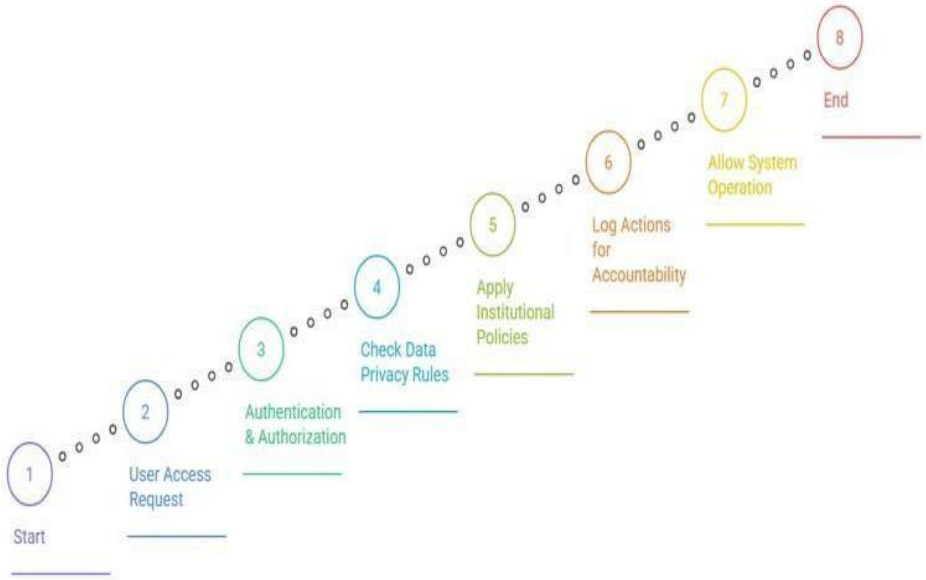

Figure 8.1 Ethical and Legal Compliance Flow

8.2 Legal aspects

The development and initial release of the Smart Classroom and Timetable Schedular entail a variety of legal matters that illustrate how the system's operation in a responsible, safe way and compliance with the law and institutional regulations have been ensured. A major element in protecting the rights of users, securing the institution's data, and preventing legal disputes that may arise because of the utilization of the system is the resolution of legal issues.

Among the legal matters that have been discussed in Smart Classroom and Timetable Schedular, one of the most important is the issue of data protection and privacy. The system stores and manages the sensitive information of the faculty, department, user's login password, and timetables. Privacy violations or misuse could be the result if these data were abused. As a measure to avoid this, Smart Classroom and Timetable Schedular has been fitted with controlled access mechanisms that only allow authorized persons into the system. Authentication and role, based authorization ensure that confidential data are accessed exclusively for legitimate academic purposes. The implementation of such a technique is in line with the data protection fundamentals normally observed by degree, granting institutions.

One more important legal dimension is related to intellectual property rights. As Smart Classroom and Timetable Schedular is an academic project from scratch, the ownership of the source code and documentation, in this case, would typically belong to the project developer or the academic institution following the prevailing academic guidelines. The developers consciously avoided using unauthorized third, party software or copyrighted material during the development process. Open, source tools and libraries were incorporated wherever necessary, and these tools adhere to their respective license terms. This guarantees that the system does not infringe software copyright laws or license agreements.

The adherence to institutional policies and academic regulations is equally a major legal requirement. Universities and colleges frequently have rules regarding staff workload, course allocation, office hours, and the use of classrooms. Smart Classroom and Timetable Schedular is capable of producing the timetable solely by taking into account the constraints and policies of the institution that have been set beforehand. As a result, the possibility of having disputes over scheduling unfairness, unhealthy workloads, or other policy breaches is minimized, thus, the legal responsibility of the institution is maintained.

The use of licensed and open, source software has therefore further verified the legal commitment of the system to public law. All the puzzle pieces of Smart Classroom and Timetable Schedular, i.e., development tools, frameworks, and platforms are made up of open, source or properly licensed software, including programming languages, database systems, and servers. Thus, the issue of software pirating or unauthorized deployment which is the root cause of the problems is legally solved and hence the software laws are obeyed.

Furthermore, the intrinsic characteristics of Smart Classroom and Timetable Schedular are accountability and traceability which are also the most important legal requisites of administrative systems. Whether it is the generation of the timetable, update, or changes, such activities can always be confirmed through the system records. Therefore, it reflects only transparency and is a great support for any external audits or internal reviews regarding scheduling conflicts.

Legally, the main pillars Smart Classroom and Timetable Schedular have been well supported and there has been no reckless gamble. By protecting the data, it ensures the privacy of the users; by acknowledging the intellectual property rights of others; by complying with the institutional policies and by employing software tools that are in line with the law, the system

solves the problem of automated timetable management in a legally acceptable way, thus, being responsible. Such behaviors of system usage are facilitated and encouraged by the system itself.

Table 8.2 Sustainability and Safety Considerations

Aspect	Area	Description	Benefit
Sustainability	Paper Reduction	Digital timetable generation and access	Environment-friendly operation
Sustainability	Resource Efficiency	Reduced administrative effort and time	Long-term operational sustainability
Sustainability	Scalability	Modular architecture	Easy future expansion
Safety	Data Security	Authentication and role-based access	Prevention of unauthorized access
Safety	System Stability	Validation and error handling	Continuous and reliable operation
Safety	Backup & Recovery	Regular database backups	Protection against data loss
Safety	Infrastructure Safety	Low resource usage and UPS support	Reduced hardware failure risk

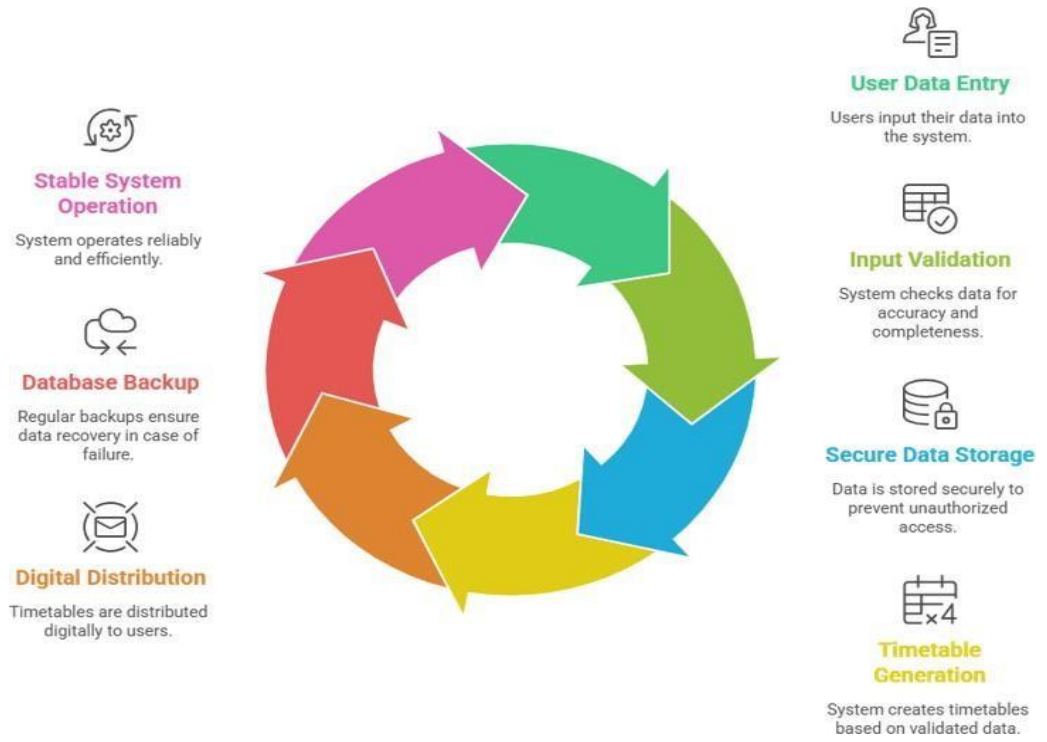

Figure 8.2 Safety and Sustainability Flow

8.3 Ethical Aspects

The creation and launching of Smart Classroom and Timetable Schedular bring about some fundamental ethical issues that revolve around fairness, transparency, accountability, and the ethical use of technology in educational institutions. When these ethical guarantors are properly dealt with, the system not only becomes a tool that upholds academic standards but also helps in gaining the trust of everyone involved such as school administrators, teaching staff, and students.

One significant ethical issue that is tackled in Smart Classroom and Timetable Schedular is the fairness of timetables allocation. Human schedulers may inadvertently bring bias, favoritism, or put an unequal workload on teachers without even realizing it. Smart Classroom and Timetable Schedular solves this problem by using rule, based scheduling logic that takes into account the constraints set beforehand like the availability of faculty, the requirements of the subjects, and the policies of the institution. Such an automated system ensures that teaching

hours are distributed fairly and human bias is kept to a minimum, thereby making the generation of schedules more objective.

Transparency forms another critical ethical dimension of the system. People using it have the right to know how timetables are crafted and on what basis certain allocations are made. Smart Classroom and Timetable Schedular ensures this transparency by rigorously adhering to a fixed set of rules and constraints while creating timetables. Also, administrators have the ability to regulate and change the parameters of the input, thus making the process of decision, making not only visible but also responsible. This level of openness is one of the ways of gaining users satisfaction with the system and along with that it lowers the chances of disagreements.

At the same time, the system, led awareness campaign about the importance of data handling cautiously is one of the points raised. From an ethical point of view, personal and institutional data should only be used for legitimate purposes. Faculty and student data that the Smart Classroom and Timetable Schedular system utilizes are only for academic scheduling. It is neither misusing data nor revealing information to unauthorized parties. Access control functions, such as, are a way to protect sensitive information storage from unauthorized users. Naturally, such measures lead to ethical data usage practices.

Moreover, There is also the issue of autonomy and human oversight of users, as discussed in another ethical paragraph. Smart Classroom and Timetable Schedular is set up to largely automate the process of timetabling but human participation is still necessary. The administrator, for example, is given the opportunity to examine, change or even completely reject the schedule. Hence, the combination of machine efficiency and human decision, making keeps the human ethical judgment in scheduling (especially when dealing with special cases).

The ethical use of technology also encompasses accuracy and accountability. Smart Classroom and Timetable Schedular has been developed to reduce mistakes as much as possible through validation checks and thorough testing. Nevertheless, the responsibility of giving the final approval to the schedules stays with the authorities of the institution. This division of responsibility is a way of ensuring that ethical accountability is upheld, and that automated outputs are checked prior to their implementation.

Moreover, Smart Classroom and Timetable Schedular is a tool that encourages ethical technology adoption by offering digital productivity as a complement to human work. Rather

than removing the duties of the administrative staff, the system is there to help them by taking over the repetitive tasks so that they can concentrate on more significant academic activities. The fact that the system plays a supportive role is in line with the ethics of technology being used for people and not against them.

8.4 Sustainability aspects

Sustainability should be a key principle when designing and developing software systems today, especially in schools where it is important that such systems be used for a long time, that they are energy, efficient, and that they can be easily adjusted. Smart Classroom and Timetable Schedular helps sustainability in a number of ways: it encourages resource efficiency, cuts down on manual work, and caters to the long, term needs of the institution.

Limiting paper consumption is a major sustainability feature on which Smart Classroom and Timetable Schedular concentrates. In the old days, the preparation and distribution of timetables would involve a great deal of printing a pile of documents, notices, and updates. Smart Classroom and Timetable Schedular by offering a digital, web, based tool for creating the timetable and allowing getting it radically decreases the need for paper, based ways. Thus it turns out to be green by reducing not only paper usage but also waste coming from it.

Besides, the solution is a lever for the sustainability of operations in the school setting. It takes a lot of human power to create timetables by hand and make changes repeatedly, especially if the changes are frequent Smart Classroom and Timetable Schedular will automatically do the task, thus it saves Time and Lessens the administrative load. By making academic operations more efficient, the institution will be able to continue college programs with less input of resources.

From a technology standpoint, Smart Classroom and Timetable Schedular runs on widely available hardware and open, source software tools, which guarantees its survival over time. Because the system does not demand high, end equipment, schools are able to keep using and managing it without the need for frequent hardware updates. This strategy lowers the generation of electronic waste and is advantageous for sustainable technology adoption.

The scalability and adaptability of the system are also very important sustainability aspects.

Smart Classroom and Timetable Schedular utilizes a modular design, meaning it can be upgraded or extended as the institution's needs grow. Without having to overhaul the whole system, you can add new features, departments, or restrictions. This flexibility is there to make sure the system does not get outdated and that it becomes an integral part of sustainable software lifecycle management.

Moreover, the system enhances human sustainability indirectly by alleviating the strain and workload of academic administrators and faculties. Since Smart Classroom and Timetable Schedular automates scheduling tasks that are both repetitive and time, consuming, the staff is thus enabled to dedicate more time to academic planning, teaching excellence, and student involvement. In this way, the working environment not only becomes healthier, but also improves the job satisfaction level.

Moreover, Smart Classroom and Timetable Schedular promotes ecologically, friendly digital habits through the centralization of timetable management. Digital updates and changes can be made and instantly reflected across the system, which greatly diminishes the necessity of communication and manual coordination. This efficient method helps to save and utilize the resources of the educational institutions.

Last but not least, using open standards and popular technologies guarantees that Smart Classroom and Timetable Schedular can be maintained and improved in the future without being tied to proprietary tools. This not only supports the economic aspect but also the technical sustainability of the system, thus making it a dependable solution for educational institutions.

8.5 Safety aspects

Ensuring safety in software systems is very important, especially for ones like Smart Classroom and Timetable Schedular that handle institutional data and support academics. Smart Classroom and Timetable Schedular has safety features built in to make sure its usage is safe, the data is protected, and the system is reliable in an educational environment.

Smart Classroom and Timetable Schedular tackles data security issues as one of its main safety features. The system manages confidential information such as faculty details, user credentials, and timetable records. To keep the data safe, the system access is controlled via authentication

methods which guarantee that only authorized users can log in and carry out operations. This way any unauthorized access is prevented, and the chances of data being misused or mistakenly changed are reduced.

System reliability and stability are other very critical safety aspects. Smart Classroom and Timetable Schedular is built on tried and tested technologies that are very stable and have fewer chances of system failures. Validation is done at various levels to ensure that only correct data is entered into the system; otherwise, it could produce incorrect timetables. The software is designed to cope with errors in a friendly manner so that it does not crash and provides service continuously.

The system also improves operational safety in that it assures timetable generation will not create any conflicts that might disrupt the academic activities. In this context, safety, among other things, is the prevention of scheduling errors such as faculty clashes, overlapping classes, or the misuse of rooms. By imposing the restrictions and rules that were set beforehand Smart Classroom and Timetable Schedular assures the safety of the timetables it produces for practical applications.

Concerning the user, Smart Classroom and Timetable Schedular guarantees safe usage of the system by providing a simple and user, friendly interface. Neatly organized navigation, helpful messages, and confirmation dialogs all combine to avert the occurrence of unintentional actions such as the accidental deletion of data or incorrect submissions. Thus, the possibility of user errors is greatly reduced, and the overall safety of the system is enhanced.

It further boosts operational safety by ensuring that the schedule generation process will not lead to any conflicts that can disrupt teaching. Here, safety is, among other things, the avoidance of scheduling errors such as faculty clashes, overlapping classes, or the misuse of rooms.

Smart Classroom and Timetable Schedular, by enforcing the restrictions and conditions that were defined beforehand, guarantees the safety of the timetables it generates for practical purposes.

For the user, Smart Classroom and Timetable Schedular will ensure the safe use of the system by offering the user a simple and friendly interface. Well, arranged navigation, useful

messages, and confirmation dialogs all work together to prevent the occurrence of unintentional actions such as the accidental deletion of data or incorrect submissions. Therefore, user mistakes are considerably less likely, and the overall system safety is increased.

Data backup and restoration also contribute significantly to data security.

It is a good measure to do regular backups of the database to avoid the loss of data in cases of unexpected network crashes or sudden events. If there are issues, the backup information might be retrieved so that the operations may stay at the level of normalcy.

In addition, Smart Classroom and Timetable Schedular aims to increase the safety of the infrastructure by being a reliable product if used on regular hardware with low consumption of resources. This brings down the risks of hardware failure, getting overheated, or crashing of the system.

Further, the system might be backed up by means of very simple power backup devices like UPS to be protected from power failures.

Chapter 9

Conclusion

The Smart Classroom and Timetable Schedular is a product of a project aimed at changing the complex and time, consuming method of creating academic timetables to a simpler and more efficient one. The project resulted in a practical solution that dealt with the major issues that everybody encounters in such a process, i.e. scheduling conflicts, manual errors, workload imbalance, and frequent revisions. Smart Classroom and Timetable Schedular is well, implemented, and its results prove that automation can make the task of timetable management in schools stronger, faster, and more dependable anymore.

One of the project's main successes was the use of rule, based scheduling logic to produce conflict, free timetables. The system, by taking into account constraints like faculty availability, subject requirements, operating hours, and classroom usage, makes sure that the schedules generated are not only correct but also feasible. The system's rigidness reduces errors and therefore solves the issues of session overlapping and workload distribution at least as far as manual scheduling methods are concerned.

One of the main factors that have contributed to the overall success of the system and its ease of use, was the decision to make it a web, based system. Smart Classroom and Timetable Schedular enables administrators and faculty members to use the system via a regular web browser without having to install any extra software, thereby guaranteeing easy accessibility. Such a step has turned the instrument into a user, friendly one and has, thus, made possible the efficient flow of communication between different departments. In addition, centralized management of timetable data not only guarantees data consistency but also greatly lessens the chances of misunderstanding.

From a technical perspective, this piece of work is a very good example of how frontend technologies, backend logic, and database management can be harmoniously combined to form a fully functional system. Employing modular architecture in Smart Classroom and Timetable Schedular not only makes the source code more comprehensible, easier to maintain, and less intimidating but, the carefully designed layout also helps in adding new features and, hence, the system can satisfy the changing needs of the college.

Furthermore, the systems utility and sustainability over time are enhanced by the choice of hardware that is generally available and the use of open-source software tools.

The evaluation and testing phases demonstrated that Smart Classroom and Timetable Schedular is stable and behaves as expected in the real world. The system managed various test scenarios featuring constraint validation, data integrity, and user interaction flows without failing. Performance testing revealed that timetable generation was completed in a moderate time frame even at medium data volume. These findings confirm the effectiveness of the system design and prove that the system can be used in real situations.

Besides technical performance, the project touches on social, ethical, legal, environmental, and safety aspects.

Smart Classroom and Timetable Schedular is designed to allocate schedules fairly and transparently, safeguard sensitive school data, decrease paper usage, and promote sustainable digital habits. By ensuring responsible automation and keeping humans in the loop, the system adheres to academic values and ethical standards.

The project experience gave the team a chance to learn software design, problem analysis, and system evaluation through play. The process of creating Smart Classroom and Timetable Schedular demanded the team to have in, depth knowledge of real, world scheduling issues and skill in converting them into logical constraints and system rules. That strengthened the team's practical programming, database design, testing, and documentation skills, thereby making a big contribution to their overall technical and professional development.

9.2 References

1. Sommerville, I., *Software Engineering*, 10th Edition, Pearson Education, 2016.
2. Pressman, R. S., and Maxim, B. R., *Software Engineering: A Practitioner's Approach*, 8th Edition, McGraw-Hill Education, 2015.
3. Oracle Corporation, *Java Platform, Standard Edition Documentation*,
   Available at: https://docs.oracle.com/javase/
   (Accessed during project development).

4. Apache Software Foundation, *Apache Tomcat Documentation*,
   Available at: https://tomcat.apache.org/
   (Accessed during project development).

5. MySQL Documentation, *MySQL 8.0 Reference Manual*,
   Available at: https://dev.mysql.com/doc/
   (Accessed during project development).

6. W3C, *HTML, CSS, and JavaScript Specifications*,
   Available at: https://www.w3.org/
   (Accessed during project development).

7. GitHub Repository, *SMART-TIMETABLE-SCHEDULER*,
   Available at: https://github.com/Pavan19047/SMART-TIMETABLE-SCHEDULER
   (Accessed for system design and implementation reference).

8. Silva, A., and Rocha, A., "Timetable Scheduling and Automation in Educational Institutions," *International Journal of Computer Applications*, Vol. 175, No. 22, 2020.

9. IEEE Computer Society, *IEEE Standard for Software Testing Documentation*, IEEE Std 829-2008.

10. ISO/IEC 25010, *Systems and Software Quality Requirements and Evaluation (SQuaRE)*, International Organization for Standardization, 2011.

9.3 Base Paper

The Smart Classroom and Timetable Schedular development draws heavily from the ideas and methods described in the chosen base paper on automated timetable scheduling for educational institutions. The base paper was a key reference that helped to understand the scheduling problems and pointed towards the design of an efficient automated solution.

The base paper mainly points out the drawbacks of traditional manual timetable making like it is very time, consuming, there is a higher chance of having scheduling conflicts, it is not flexible, and it is very difficult to consider different constraints simultaneously. It calls for automation to handle situations where the academic environment is very complex, and at the same time, faculty availability, subject requirements, and institutional policies must be taken into account.

These issues fit well with the ones that the real, world Smart Classroom and Timetable Schedular system is solving. The base paper introduces a major change in the way of constraint, based scheduling where a set of predefined rules are used to generate a timetable without any conflicts. The authors of the paper have discussed this and hence, Smart Classroom and Timetable Schedular is deeply influenced by this idea. They have taken a similar rule, based methodology to check faculty availability, time slot allocation, and classroom usage before posting the final timetable. This method increases the reliability and uniformity of the generated timetable.

While the base paper is largely theoretical and algorithmic in its approach to timetable automation, Smart Classroom and Timetable Schedular takes these concepts and transforms them into a tangible, web, based product. The system doesnt just stop at authentication and database integration. It also features a friendly and easy, to, navigate interface, making it a viable solution for schools. By doing this practical extension, the authors have successfully linked academic research with practical application.

Another key takeaway from the base paper is the emphasis on scalability and adaptability. The paper advocates that automated scheduling systems should have enough flexibility to support changes within the institution. Smart Classroom and Timetable Schedular is basically a reflection of this idea as the application of a modular design makes it possible for the program to be easily upgraded, enhanced, and scaled without a complete overhaul.

Furthermore, the base paper played a role in shaping the projects emphasis on fairness and transparency of timetable distribution. A scheduling system that is automated, by nature, will lessen human interaction and bias thus, bringing planning to be ethical and well, balanced.

9.4 Appendix

The appendix section contains further technical and supporting information regarding the Smart Classroom and Timetable. The materials included in this section clarify the internal functioning of the system, details of implementation, and the testing environment. These topics have been dealt with in the appendix in order not to interfere with the main chapters' content flow. The appendix is a useful reference for reviewers who want to delve deeper into the technical side of the project.

Appendix A: Sample Source Code

This appendix displays some selected and partial code snippets which illustrate the main functionality of the system. The complete source code is stored separately in the project repository, whereas only representative parts are shown here for the sake of clarity and in compliance with academic requirements.

A.1 Timetable Generation Logic (Partial Code)

```
public class TimetableGenerator {

public void generateTimetable(List<Subject> subjects, List<TimeSlot> slots) {
for (Subject subject : subjects) {
for (TimeSlot slot : slots) {
if (slot.isAvailable()) {
subject.assignSlot(slot);
slot.setAvailable(false);
break;
}}
}
}
}
```

A.2 Database Connection Module

```
public class DBConnection {

public static Connection getConnection() {
Connection con = null;
try {
Class.forName("com.mysql.cj.jdbc.Driver");
con = DriverManager.getConnection("jdbc:mysql://localhost:3306/timeforge",
      "root","password"
      );
  } catch (Exception e) {
      e.printStackTrace();
  }
```

```
    return con;
}
}
```

Appendix B: Database Design Overview

The Smart Classroom and Timetable Schedular database is designed with a relational model to keep data consistent and reliable. There are several main tables that are part of the system:

User Table: Contains login details and roles of access.

Faculty Table: Contains information about faculties and their availability.

Subject Table: Holds department and subject data.

Time Slot Table: Contains information about working days and sessions.

Appendix C: System Screenshots Description

The following screenshots were captured during system execution to demonstrate key functionalities:

- Login interface
- Faculty and subject data entry screens
- Timetable generation page
- Generated timetable display
- (Screenshots can be attached separately as per submission guidelines.)

Appendix D: Software and Tools Used

- Programming Language: Java
- Frontend Technologies: HTML, CSS, JavaScript, JSP
- Backend: Java Servlets
- Database: MySQL
- Server: Apache Tomcat

- IDE: Eclipse
- Version Control: Git

Appendix E: Test Environment

- Operating System: Windows / Linux
- Browser: Google Chrome / Mozilla Firefox

Hardware: Standard desktop or laptop meeting minimum requirements

Report Similarity Check

turnitin Page 2 of 8 - AI Writing Overview Submission ID trn:oid:::1:3501793162

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

turnitin Page 2 of 9 - Integrity Overview Submission ID trn:oid:::1:3501793162

3% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

Match Groups

- **8** Not Cited or Quoted 3%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

2% Internet sources
2% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Research Paper Submission

Fourth International Conference on Innovations in Software Architecture and Computational Systems : Submission (2) has been created. Inbox

Summarize this email

Microsoft CMT 8:57 AM
to me

Hello,

The following submission has been created.

Track Name: ISACS2026

Paper ID: 2

Paper Title: TimeForge-Smart Classroom and Timetable Scheduler

Abstract:
One of the most perennial logistical problems in higher education is the effective allocation of its very limited academic resources such as time, space, and teaching members. Manual scheduling, apart from being a very time, consuming task, is always accompanied by a large number of mistakes that result in only partial solutions to problems such as overlapping of resource, uneven distribution of faculty workload, and inefficient utilization of classrooms.
In this paper, we offer the